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1 INTRODUCTION

This document presents the findings of a conceptual study regarding the high level hyperloop concept, with basis on the work of the Hyperloop Development Program (HDP). This work has been carried out with a focus on passenger transport, with considerations coming from the Cargo Hyperloop workgroup from the HDP.

Three different hyperloop concepts have been evaluated, a Small (S), Medium (M) and Large (S), referring to the dimensions of the system. These different concepts have been evaluated in different ways to identify the most promising direction and proceed with a recommendation for development.

This document presents an overview of this process.

2 CONTEXT

On a global scale, the increasing demand for energy, services, home heating, internet, transport, etc., has put us in a position in which action towards reducing this energy use needs to be taken to ensure our planet and our own wellbeing.

We already see effects of this change in the shape of deforestation, the melting of ice caps, wildfires, draughts, colder summers, warmer winters, etc. About 23% of the entire global emissions come from the transport sector. Basically, almost 1 out of every 4 grams of CO₂.

The issue is that the current transport infrastructure is unable to function without negatively affecting the climate of the planet. Furthermore, it is estimated that transportation demand will triple by 2050. Upwards of €50 trillion will be necessary to accommodate the growing demand. This should not happen in legacy-only infrastructure.

Hyperloop is a modality under development that will play an important role in alleviating transport congestion, preventing further climate degradation and creating a future-proof mobility landscape.

The vision is a global, high-speed network of sustainable transport.



Figure 1. European hyperloop network for passengers and cargo

To further define the vision for passenger transport, 3 main research blocks were identified and are described below.

- **Product Market fit. (Chapter 5.1)**
- This section deals with fundamental questions about hyperloop in the transportation market. It explores how current modes of transports operate and how could hyperloop compare. Furthermore, it provides insight into the end-user wishes and needs.
- **Passenger safety, requirements and regulations (Chapter 5.2)**
- This section looks into the different regulations, requirements and safety considerations that exist in other modes of transport such as high speed rail, aviation, regular rail and even rollercoaster

regulations that could be applicable to hyperloop. Furthermore, it presents an overview of different emergency scenarios and possible mitigation strategies.

- **Product design and passenger experience. (Chapter 5.3)**
- This section compares three potential scenarios for small, medium and large hyperloop systems; dubbed hyperloop S, M and L respectively. It provides an overview of the different layouts for passenger accommodation, strengths and weaknesses and presents a comprehensive tradeoff process.

3 DOCUMENT GOALS

The goals of the document are the following

- Establish service characteristics
 - **Service characteristics.** Key to make hyperloop service competitive. Based on market requirements
- Establish a concept hyperloop service
 - **High level passenger journey.** Key to understanding the passenger touchpoints.
 - **General Concept of Operations diagram.** Key to understanding the operations of the whole system
 - **Luggage handling.** Key parameter for sizing of the vehicles and definition of the passenger journey
 - **Inclusivity.** Design for People with Reduced Mobility
- Establish a concept hyperloop product
 - **Vehicle concept.** Cross section and capacity are key for costing and throughput calculations
 - **Safety concept.** Key parameter to place emergency systems and processes throughout the hyperloop

The findings from the research and the results from the tradeoff process are summarized in **Concept description (Chapter 6)**.

4 HYPERLOOP ESSENTIALS

This section provides an overview of the basic hyperloop system, to create an understanding of the elements that compose it and how they interact with each other.

4.1 SYSTEM ARCHITECTURE CONCEPT

The hyperloop is a mode of transport for large volumes of passengers and cargo. Some high-level systems have already been defined and are considered to be hyperloop's essential systems. Without too much detail about which specific solutions are chosen, the System Architecture Concept (depicted below) gives a good overall idea of how hyperloop operates.

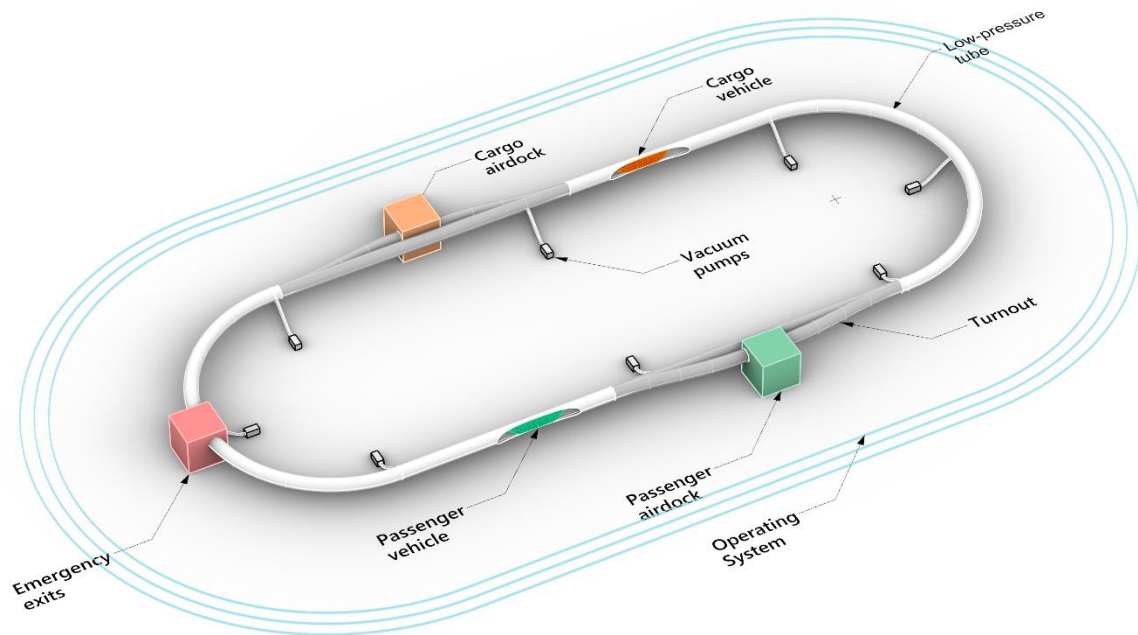


Figure 2. Hyperloop System Architecture Concept

The high-level systems are the following

- Low-pressure tube / guideway
 - It provides structure, general alignment, and holds the low-pressure environment.
- Turnout
 - Segment of the infrastructure that allows the low-pressure tubes to be bifurcated without requiring the vehicles traveling through it to reduce their speed
- Passenger vehicle
 - Safely and comfortably transports passengers and their belongings
- Passenger airdock
 - Interface between the low-pressure tube and the ambient. It allows for the transfer of passengers between the vehicles and the ambient.

- Emergency exits
 - Provide safe passage to passengers from the vehicle, through the tube, to the external environment.
- Cargo vehicle
 - Safely transport cargo
- Cargo airdock
 - Interface between the low-pressure tube and the ambient. It allows for the transfer of cargo between the vehicles and the ambient.
- Vacuum pump
 - Creates the low-pressure environment within the tube and assists with maintaining that low pressure by offsetting the infrastructure leakage.
- Operating system.
 - It autonomously controls all other systems within the hyperloop system.

Hyperloop is unique in various ways, even though all essential technologies powering it are already proven. The key advantages of hyperloop over other modes are depicted in the image below.

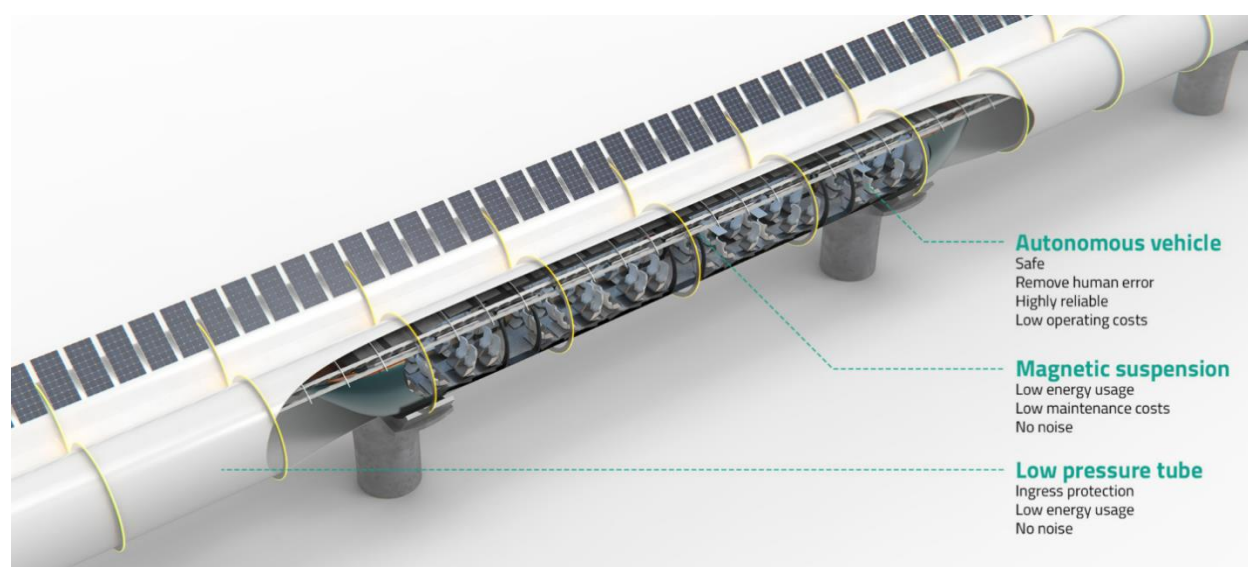


Figure 3. Hyperloop system overview

4.2 NETWORK

The vehicles travel through a network of tubes, much like a highway. In this network, vehicles travel at high speeds and can change from one tube to another without having to slow down by using the high-speed switch. The high-speed switch is an essential part of hyperloop technology, since it enables the network. Without it, the hyperloop would have to be a point-to-point system.

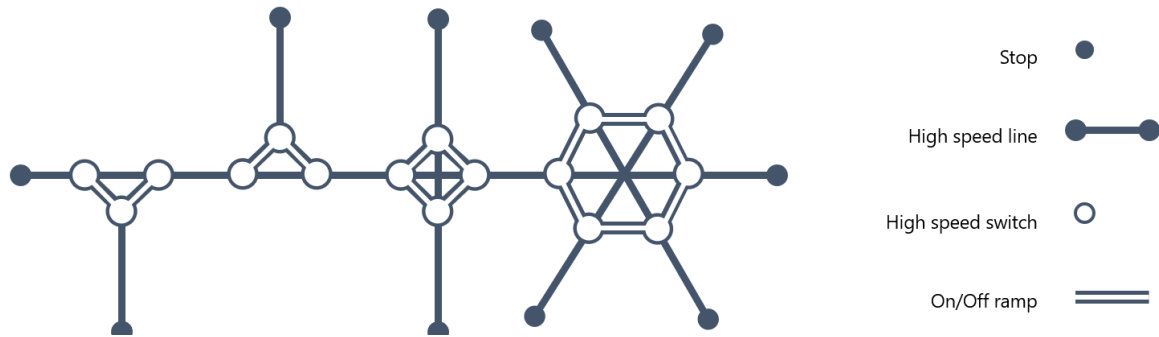


Figure 4. Hyperloop network topology

The network has 4 main components that are shown in the above diagram.

- Stops. Passenger or Cargo hubs, final destination of the trips.
- High speed line. Main part of the network, undisrupted smooth sections that allow vehicles to travel at high speeds.
- High speed switch. Switching mechanism that allows vehicles to change from one tube to another.
- On/Off ramp. Like a highway, vehicles must accelerate before joining the main line. The ramps are additional segments of infrastructure that serve the purpose of accelerating and decelerating the vehicles.

The hyperloop network is expected to house a combination passengers and cargo transport, within one single infrastructure. Passenger-only or Cargo-only corridors can exist, but it is foreseen that the network's ability to transport either or both in the same infrastructure would be extremely beneficial.

Hyperloop would use different vehicles for cargo and for passengers, as well as different hubs for the loading and unloading of cargo, or the embarking and disembarking of passengers. An overview of how these processes is depicted below.

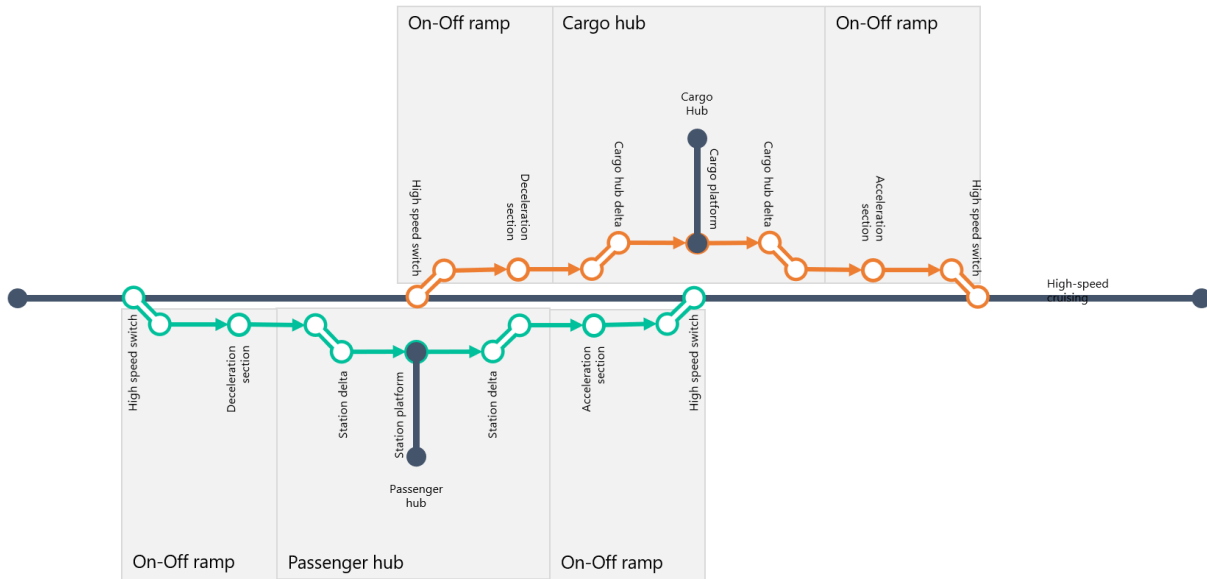


Figure 5. Passengers and cargo access processes.

4.3 SPECIFICATIONS

An overview of specifications is presented in the following table.

SYSTEM SPECIFICATIONS	VALUE
TOP SPEED:	1000 km/h
CRUISE SPEED:	700 km/h
CAPACITY/THROUGHPUT	20.000 Pax/hr/dir (at 700 km/h) (potentially multiplied with platooning)
LONG. ACCELERATION	0.15 g (average), 0.2 g (peak)
VERTICAL ACCELERATION (CREST)	0.06 g
VERTICAL ACCELERATION (TROUGH)	0.12 g
MAX. LONGITUDINAL JERK	0.1 g/s
MIN. CORNER RADIUS:	200 m
CORNER ACCELERATION:	0.5 g (made possible by high degree canting.)
MAX. LATERAL JERK	0.05 g/s
SWITCHING:	Up to 700 km/h
HEADWAY AT CRUISE:	12 sec, approx. 1.2km at 700 km/h
MAXIMUM SLOPE:	10%
RIGHT OF WAY:	8m width (Without service road)
TURNAROUND TIME:	3 minutes
EMERGENCY DECELERATION RATE	Up to 1.2 g

5 BUSINESS OPPORTUNITY DEFINITION

5.1 PRODUCT-MARKET FIT

Hyperloop is a part of a passenger transport system, providing high speed services over a wide range of distances. Hyperloop services tap into the existing passenger transport demand, providing an additional transport option complementing and substituting current transport mode.

The attractiveness of hyperloop services varies depending on various demand drivers, such as passenger needs, expectations and preferences, but also their socioeconomic and behavioral profiles. Defining the potential for modal shift from other modes of transportation to hyperloop services requires in-depth understanding of these drivers, as factors impacting passenger decisions and modal choices. There stands the scope of this study positioning hyperloop as a passenger modality next to car, high-speed rail and aviation.

Car is considered an autonomous and flexible mode of transport. Regardless of its low utilization rate and efficiency, the motorized mode of transport is the preferred one for commuters (Juhász & Golias, 2017). Besides its convenience and flexibility, owning a car (private mode of transportation) enables combining trips of multiple purposes (i.e., commuting, leisure, supply route etc.). Even though road transport is subject to high levels of taxations, its share is rather inelastic (Pastori, et al., 2018). As for high-speed rail, Spain, France, Germany and Italy operate the most extensive high-speed rail networks. Even though there are routes (e.g. Paris-Lyon, Madrid-Seville) where high-speed rail is dominant, at European scale the modal share of international rail amounts to 7 percent and of inland rail to 7.8 percent as for 2018 (RailFreight, 2021). Aviation facilitates inter-regional and international connectivity by means of domestic, short, medium to long-haul international flights. The demand of domestic and short (~ less than 3 hours flight time) flights can theoretically be covered by public transport services (rail, bus or high-speed rail services) or car. Hyperloop as a high-speed modality is expected to offer reduced travel time and to supplement other modalities (mainly high-speed rail and aviation) or to compete with them.

Research is underway to understand the share of hyperloop when it is positioned next to other modalities. Hence, the objectives of this study lie on four main pillars: to define key modal shift criteria for passengers to choose hyperloop over other modes of transport; to identify the differences between modal shift criteria for different market segments by distance and by purpose; to differentiate the modal shift functions for the main hyperloop passenger market segments and to design a modal split model for hyperloop.

5.1.1 SEGMENTING THE MARKET

Transport networks support many passenger movements. These passengers represent one part of the market (the other being cargo) and can be segmented into four broad categories to obtain insights into a passenger's profile. For example, their needs (for a transport service), their means (ability to travel), and their attitudes (preferences and perceptions). That gives five core market segments: by distance, by purpose, by frequency, by demographics and by behavioral attitudes. A summary of these market segments, along with their descriptions and examples are outlined at



Table 1 and
Table 2.

Table 1 Market segments by distance

User	Geography	Distance (kilometers)	Example of trip
Very short distance	Urban trips	0 – 15	Within Amsterdam
Short distance	Suburban trips	15 – 60	Dusseldorf-Cologne
Medium distance	Regional trips	60 – 150	Hanover-Bielefeld
	National or short-distance international trips	150 – 300	Amsterdam – Cologne
	Long distance international trips	300 – 600	Zaragoza-Madrid
Long distance	Long distance international trips	>600	Amsterdam-Berlin

Table 2 Market segment by purpose-work related trip purposes: work related and leisure

Market segment	User	Description	Example of trip
Work-related	Commuter	A passenger who travels regularly for shorter distances to work	Travel to-and-from an office for work
	Business Traveler	A passenger who travels irregularly (or on an at-needs basis) for longer distances for work-related purposes	Travel to-and-from a trade fair for work
Leisure	Health	A passenger who travels to services related to their health	Travel to-and-from a practitioner for health services
	Education	A passenger who travels to services related to their education	Travel to-and-from school for education
	Recreation	A passenger who travels to a destination for their recreation and enjoyment	Travel to-and-from a sports park for wellbeing
	Tourism		Travel to-and-from an attraction for a holiday
	Personal	A passenger who travels to a destination for personal reasons	Travel to-and-from a destination to visit family and friends

5.1.2 TRAVEL EXPERIENCE AND TRAVELERS' NEEDS

Numerous studies have been conducted to measure the quality of various passenger transport services, and to identify passenger needs, expectations and priorities. The common denominator of the results of this research is a set of passenger experience descriptors, or service parameters which impact passenger experience and satisfaction from the journey. Some of these descriptors are so important that they impact the modal choice decisions. These descriptors will be later called passenger experience differentiators.

5.1.2.1 TRAVEL EXPERIENCE

Passenger experience is an all-encompassing term describing subjective perception of a journey, directly related to transport service quality. In all transport modes, transport service quality is measured primarily via customer satisfaction surveys and customer expectation surveys. The gap between these two gives a valuable insight into the required changes in transport services and their prioritization. The number and nature of transport service descriptors used in these surveys differ on a case-by-case basis, but there is a number of common denominators:

- **Operational experience**, including transit time, reliability and punctuality of service. Transit time takes into account the actual in-vehicle time plus the time spent for baggage controls, identification, check-in, waiting time as well as the minimum time needed for handling baggage and reaching the gate.
- **On-board experience**, including availability of seats and amount of space per passenger (level of crowding), cleanliness of vehicles, availability of sanitary facilities and their cleanliness, safety and personal security, additional services such as Wi-Fi or meals, availability of space for luggage. The on-board experience is perceived by passengers as the comfort of using a given transport mode.
- **Access/egress experience**, including access/egress time (a measure of accessibility and connectivity of stations), frequency of departures, information provision at stations, security at the station, availability of station facilities, quality of station environment. The access/egress experience is perceived by passengers as the convenience (Andersoni, Condryi, Findlayi, Brage-Ardaoi, & Lii, 2012) of using a given transport mode.
- **Purchase experience**, including pricing, effort needed to purchase tickets, compensation for disruptions
- **Sustainability experience**, focused on perceived environmental impact of a journey.
- **Capacity**. The system must have enough seating space to fulfil the travel demand

Table 3: Passenger experience descriptors

Category	Descriptor	Definition
Operational experience	Transit time	In-vehicle time plus the time spent for baggage controls, identification, check-in, waiting time as well as the minimum time needed for handling baggage and reaching the gate
	Reliability/punctuality	How closely the actual service matches the route timetable
	Capacity	Passenger capacity offered by the system during a given period of time. For example, passengers per hour, or passengers per year.
Comfort	Vehicle condition	The physical and mechanical condition of vehicles, including frequency of breakdowns
	Vehicle environment	How comfortable the journey is regarding to seat spaciousness, noise levels, driver handling, air conditioning
	Availability of seats	How comfortable the journey is regarding to seats availability (crowding)
	Aesthetics	Appeal of vehicles, stations and waiting areas to users' senses
	Safety and security	How safe from traffic accidents passengers feel during the journey as well as personal security
	Vehicle facilities	Availability of sanitary facilities on-board and sufficient space available on-board for passengers' luggage
	On-board services	Are additional services available on-board: Wi-Fi, mobile signal, meals, staff support
Convenience	Frequency of service	How often the service operates during a given period
	Information provision	How much information is provided about routes, interchanges and delays before and during the trip
	Ease of transfers/interchanges	How simple transport connections are, including time spent waiting for other modes of transportation
	Access/egress time	Access/egress time by other means of transport
	Stations facilities	How easy it is to access and use stations, availability of basic facilities (ticket purchase, sanitary facilities, accessibility facilities), is there staff available for support
	Station services	What additional services are available at stations
Purchase experience	Price	The monetary cost of travel
	Ticket purchase	How easy it is to buy a ticket before a trip
	Compensation for disruptions	How easy it is to claim compensation for disruptions
Sustainability experience	Carbon footprint of a journey	How environmentally friendly is the journey
	Energy mix of a journey	How much of renewable energy sources are used during the journey



5.1.2.2 TRAVELER’S NEEDS

The passenger experience descriptors impact the emotional response of the passenger in different ways. Some of them are required to be of a high standard to meet the basic needs of passengers (dissatisfiers), others impact customer satisfaction from a journey (satisfiers). Satisfiers raise the level of customer satisfaction but dissatisfiers are the most important factors impacting the modal choice.

The basic needs of passengers are focused on safety/security and reliability. Safety and security are prerequisites in the decision for choosing a travel mode. If potential customers perceive the journey by a certain transport mode to be unsafe, they will avoid it. Moreover, if the safety expectation is not met, passengers are willing to abandon the journey if an alternative transport mode is not available. In this sense, safety is an absolute prerequisite for a mode of transportation to be considered as a viable option.

Reliability is broadly recognized as a desirable transport network attribute. However, in a review of relevant literature, it was found that transport reliability was defined in a number of different ways. In this study, a transport service is considered reliable as long as travel times are consistent, even if the service underperforms due to regularly slow speeds caused by traffic congestion. In this context, from the user perspective, reliability can be defined as the ability of the transport service to provide the expected level of service quality, upon which users have organized their activities. The unexpected disruption to personal and commercial schedules resulting from unreliable transport services has a significant impact on perception of transport services. In planning a journey, the user has not just to consider the expected average travel time but also its variability and uncertainty. If the traveler wants to reduce the risk of disruptions to their activities at the destination, they will have to allow more time than the mean travel time (safety margin or buffer). The need to build in these buffers can be costly in terms of waiting times during travel or costs in the intended activity if the buffer time is not sufficient, creating scheduling problems such as missed or delayed connections or appointments.

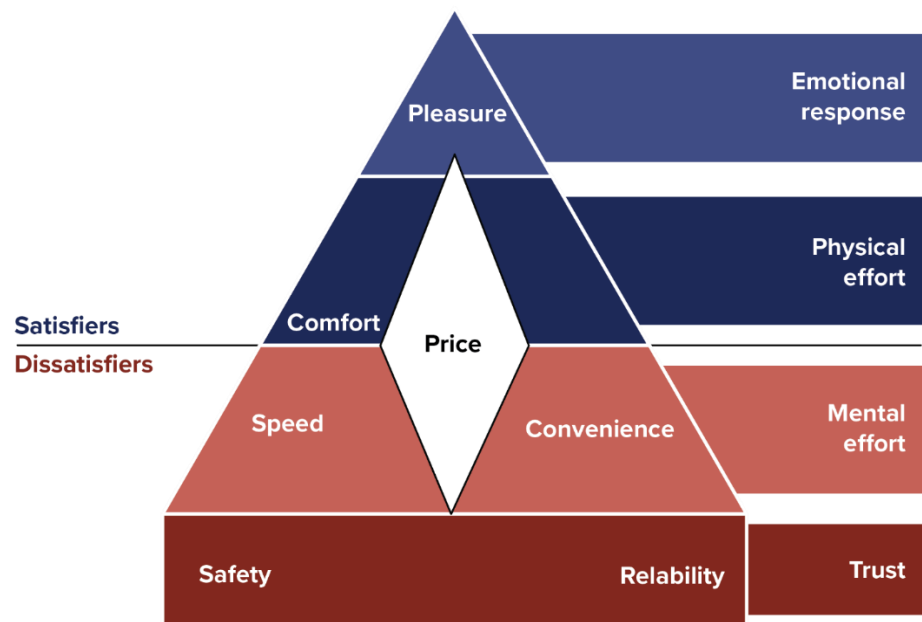


Figure 6 Pyramid of customer needs (Hagen & Bron, 2014)

Perception of safety is built on trust generated by external information and long-term community experience. Reliability is also built on trust but in a short-term perspective. As long as the passenger is given information about what they can expect and this information proves to be reliable and trustworthy during their journey, their trust is reaffirmed and maintained.

The next level of passenger needs, and expectations is related to travel time, convenience and price. Each customer makes a judgement between these three factors when planning a journey. Speed is the principal customer need, but the passenger wants the trip to be easy, convenient, with and with little hassle. These two factors against the ticket price define how a transport option is perceived in terms of its value for money.

Satisfiers are comfort (sheltered waiting, sitting areas, comfortable seats, opportunity for other activities on the train) and pleasure. All passengers expect a certain level of comfort as their threshold for a decent transportation service, and if their expectations are not met, will decide to choose another alternative (if available). Above this individual comfort threshold, there is a trade-off between comfort and price - some customers are willing to pay a premium for increased comfort (and pleasure), some on the contrary, are willing to accept lower levels of comfort and pleasure to minimize monetary cost of a journey. The pleasure is enhanced by architecture, design, light, smell and music, but also personal values such as sustainability and equality and can influence the satisfaction from the passenger experience.

Passenger experience differentiators are defined as passenger experience descriptors which are most likely to impact modal choice in passenger transport. The most important criteria for transport mode selection are related to safety, reliability, speed, price, convenience, and comfort. As illustrated in **Error! Reference source not found.**, all these categories encompass a range of descriptors, with a varying impact on overall perception of passenger experience in a given category. For example, on-board services have lower impact on comfort perception than availability of seats. The literature review revealed that the descriptors listed below are the most important for assessment of a passenger transport option according to the modal choice criteria and therefore constitute passenger experience differentiators.

Table 4: Modal choice criteria and passenger experience differentiators

Criteria	Differentiators	Definition
Prerequisites	Safety and security	How safe from traffic accidents passengers feel during the journey as well as personal security
Operational experience	Reliability/punctuality	How closely the actual service matches the route timetable
	Transit time/speed	In-vehicle time plus the time spent for baggage controls, identification, check-in, waiting time as well as the minimum time needed for handling baggage and reaching the gate
Convenience	Frequency of service	How often the service operates during a given period

	Ease of transfers/interchanges	How simple transport connections are, including time spent waiting
	Access/egress time	Access/egress time by other means of transport, including time spent waiting
Comfort	Vehicle environment	How comfortable the journey is regarding to seat spaciousness, noise levels, driver handling, air conditioning
	Availability of seats	How comfortable the journey is regarding to seats availability (crowding)
	Vehicle facilities	Availability of sanitary facilities on-board
Purchase experience	Price	The monetary cost of travel

In the literature, transit time, price and convenience (as in access/egress time and frequency of services) are considered the most important decision-making criteria for modal choice (sources). Although, as mentioned in the previous sections, reliability has been found to be one of the most important attributes of transportation systems, from the modal choice perspective it is sometimes considered as extremely important, and sometimes falls behind other differentiators.

The recent study on modal shift criteria in Europe (Pastori, et al., 2018) puts reliability on the fourth place after price, transit time, and frequency. This is consistent with the results of the UK research on reasons for not using rail services (Populus, 2020). The stated preference surveys conducted over the years give mixed results, with some suggesting that passengers value reliability more than transit time (Gov.uk, 2021)¹ and some that less (Jong & Kouwenhoven, 2019)². Definitely, reliability ranks highest in terms of priorities and expectations of public transport users (Populus, 2020)³.

These differences are to a large extent the result of varying definitions and measurement methodologies of reliability. Reliability tends to be attributed lower importance when it's considered only from operational perspective, as transit time variability at the route or line-level. When the passenger experience is considered, and the research account for the reliability of the entire journey, the results tend to report very high importance (Wood, 2013), consistent with the pyramid of customer's needs (**Error! Reference source not found.**).

Also, the perception of reliability as a part of the passenger experience differs per user and between users and non-users. For example, the reasons that non-users gave for not travelling more by rail list

² Time Use and Values of Time and Reliability in the Netherlands, International Transport Forum, OECD, 2019

³ Rail Passengers' Priorities for Improvement 2020, Populus, 2020

“inconvenience” as the most important factor. In this case, the term “inconvenience” covers a general perception of disutility of rail transportation, including uncertainty and risk related to potential disruptions of the whole journey. Other sources also suggest that car users tend to perceive public transport journey as much longer, unreliable and difficult than it actually is (Jacob, van Exel, & Rietveld, 2010).

The importance of reliability in modal choice is therefore difficult to define unanimously based on available literature. However, there is little doubt that the hypothesis formulated in the pyramid of customers need is valid, and reliability plays a crucial role in the decision-making process of planning a journey. It is said that reliability “has become a victim of its own success” (Jeekel, 2010). According to International Transport Forum report, the more reliability improves the more behaviors are based on the assumption of good reliability leading to lifestyle and work choices which are partly irreversible, and which make them more vulnerable to unreliability. In other words, the more reliable transport services are, the more this reliability is taken for granted and depended on. For example, passenger movements, both for business and social purposes, have become more complex with changing patterns of employment, increased disposable income, recreational choices and leisure time. These diverse and geographically spread activities have led to more intensive use of transport, bringing greater dependence on transport to be reliable so that delays do not cascade through the busy schedule of activities. From the modal choice perspective, it is therefore essential to consider reliability as a critical decision-making factor, even if in the research it is sometimes disguised under other terms, narrowed down to punctuality or taken for granted by the respondents.

The summary of the importance of modal choice criteria and passenger experience differentiators is presented below.

Criteria	Differentiators	Relative importance of passenger differentiators, scale 1 to 5 (1 – lowest importance, 5 – highest importance)				
		1	2	3	4	5
Operational experience	Reliability/punctuality					
	Transit time/speed					
Convenience	Frequency of service					
	Ease of transfers/interchanges					
	Access/egress time					
Comfort	Vehicle environment					
	Availability of seats					
	Vehicle facilities					
Purchase experience	Price					

Figure 7: Relative importance of passenger differentiators, scale 1(low) to 5(high)

5.1.3 HYPERLOOP COMPARED TO OTHER MODALITIES

The travelers' behavior is a complicated decision-making process. The factors influencing travelers' mode selection revolve around the availability of relevant infrastructure (availability of route, car ownership), mode characteristics (transit time, comfort, convenience, ticket fares) the social environment (demographics), attitudes and perceptions (sustainability, perceptions, habits). Given the customers' needs and passengers' experience descriptors as shown in



Table 3 and Figure 6 we conducted a comparison of hyperloop with other modalities based on a two-step approach:

- a multi-criteria analysis for aviation, long-distance rail, regional rail and hyperloop,
- modal choice modelling for different passenger market segments.

Each modality serves a different market segment. That said, even if regional rail scores better than other modalities in specific experience differentiators that does not mean that it is suitable for all types of trips. That means that it operates efficiently within a specific market segment.

With regards to hyperloop, we expect that it will be as comfortable as technically possible that can be. It is expected to be well connected with the existing modalities, allowing for an efficient first and last mile transport. Comparing to the other modes, hyperloop does not perform well price-wise due to uncertainty. **Error! Reference source not found.** illustrates the relative comparison of hyperloop towards the other modalities.

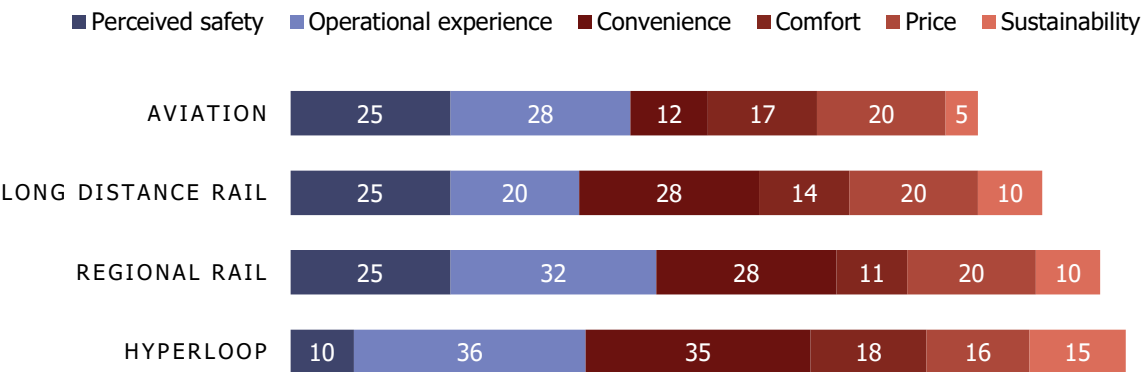


Figure 8 Hyperloop compared to other modalities

The minimum distance considered to be as viable for a hyperloop connection is that of about 15 km. Therefore, the category of very short trips (urban trips) has been excluded from this analysis. Our analysis in the context of suburban trips showed that on average 89.74% of these trips may be covered by car and the remaining percentage by hyperloop and rail services. With or without the presence of hyperloop car is an attractive alternative for inland trips (origin and destination within the same country) and short international trips. The flexibility and convenience of the private car results in it having an inelastic share. A strong preference towards car applies for regional trips: 85% of the transport demand is covered by car. The longer the routes' length becomes the higher the share for hyperloop and aviation. For (long distance) national trips, a percentage of 40% of the transport demand would potentially be covered by rail and hyperloop. Aviation captures about 7% of the market share leaving half of the demand (52.31%) to be covered by car. Eventually, in long trips with the presence of hyperloop, car loses its market share. Aviation and hyperloop provide time-wise better services and thus take most of the transport demand's share. High speed rail in some cases (e.g., Berlin-Amsterdam) cannot maintain high speed due to the extended number of stops. This type of connections serves mostly as supply for domestic trips. Hyperloop

and aviation would potentially supplement the existing corridors with capacity intending to capture the international travel, freeing capacity for the domestic trips. **Error! Reference source not found.** shows an overview of the results of the models per market segment together with the current modal split as found in the literature.

Table 5 Modal split before and after (theoretical) the introduction of hyperloop

Distance market segment	Route	Current state	Source	Car-aviation-rail / hyperloop	Deviation from current state ⁴
Suburban	Dusseldorf-Cologne	Car: 72% Rail: 26% Bike: 3%	Düsseldorfer Verkehrsforum (2018).	Car: 90% Rail / hyperloop: 10% Aviation: n.a.	Car: +18% Rail / hyperloop: -16%
Regional	Hanover-Bielefeld	Car: 85% Rail: 10% Air and coach: 5% (Long distance inland travel: >100km)	Nordenholz, Winkler, & Knörr (2017)	Car: 85% Rail / hyperloop: 15% Aviation: n.a.	Car: 0% Rail / hyperloop: +5%
National	Zaragoza-Madrid	Car-driver: 26.11% Car-passenger: 7.52% Bus: 25.22% High-speed rail: 33.19% Plane: 7.96%	Román, Espino, & Martín (2010)	Car: 52.31% Rail / hyperloop: 40.63% Aviation: 7.06%	Car: +18.68% (bus was not considered as an option) Rail / hyperloop: +7.44 Aviation: +0.9%
Short distance international	Amsterdam-Cologne	Car: 60% Bus: 10% Plane: 5% Rail: 25%	Brietzke, de Keizer, & van Oort (2015)	Car: 68% Rail / hyperloop: 32% Aviation: 0%	Car: +8% Rail / hyperloop: +7% Aviation: -5%
Long distance international	Amsterdam-Berlin	Road: 2% Rail: 9% Aviation: 90%	van Buuren, Rijniers, & Koopman (2018)	Car: 1.2% Rail / hyperloop: 54.6% Aviation: 44.2%	Car: -0.8% Rail / hyperloop: +45.6% Aviation: 45.8%

5.1.4 MARKET POTENTIAL FOR HYPERLOOP

At very short distances (suburban), the attractiveness of hyperloop is expected to be as competent as of the regular train journeys, so there's no clear advantage of introducing hyperloop. At short distances (suburban trips), travel time by hyperloop is expected to be better to regular train journeys, however accessibility may make up for that time saved. In the context of this study, car still dominates this market segment. It is difficult to compare against car, since the experience of public and private transport is very different. More research is recommended in this specific segment.

At national trips where a high-speed transport service is available hyperloop is an attractive solution for work-related trips (42%). At leisure long-distance national trips, about one out of five trips (24%) is expected to be performed by hyperloop. At short-distance international trips, hyperloop takes one third of

⁴ Comparison between: car before and after the introduction of hyperloop, rail with rail / hyperloop, aviation before and after the introduction of hyperloop.



the transport demand (35% of work-related trips, 31% of leisure trips). For medium distance national trips and short distance international trips the attractiveness of hyperloop depends on the availability of high-speed rail services. Where high speed rail services are not available, hyperloop provides a very attractive high speed transport option. Where the high-speed rail services are already in place, there’s no clear advantage of hyperloop and introducing hyperloop services in direct competition with high-speed rail should be avoided. This distance range should be considered an important market for hyperloop, but with careful case by case evaluation of potential hyperloop routes.

For the long-distance routes where a high-speed transport option is not available today, hyperloop complements the existing transport modes. The added capacity would serve as a supply for international trips. Long-distance work-related trips is expected to be the market segment with the most potential for hyperloop (63% of work-related trips, 46% of leisure trips). At long distances, national and international, hyperloop has clear advantage over rail and aviation, and this market should be considered a priority market for hyperloop. **Error! Reference source not found.** summarizes the market share for hyperloop per market segment by distance and trip purpose.

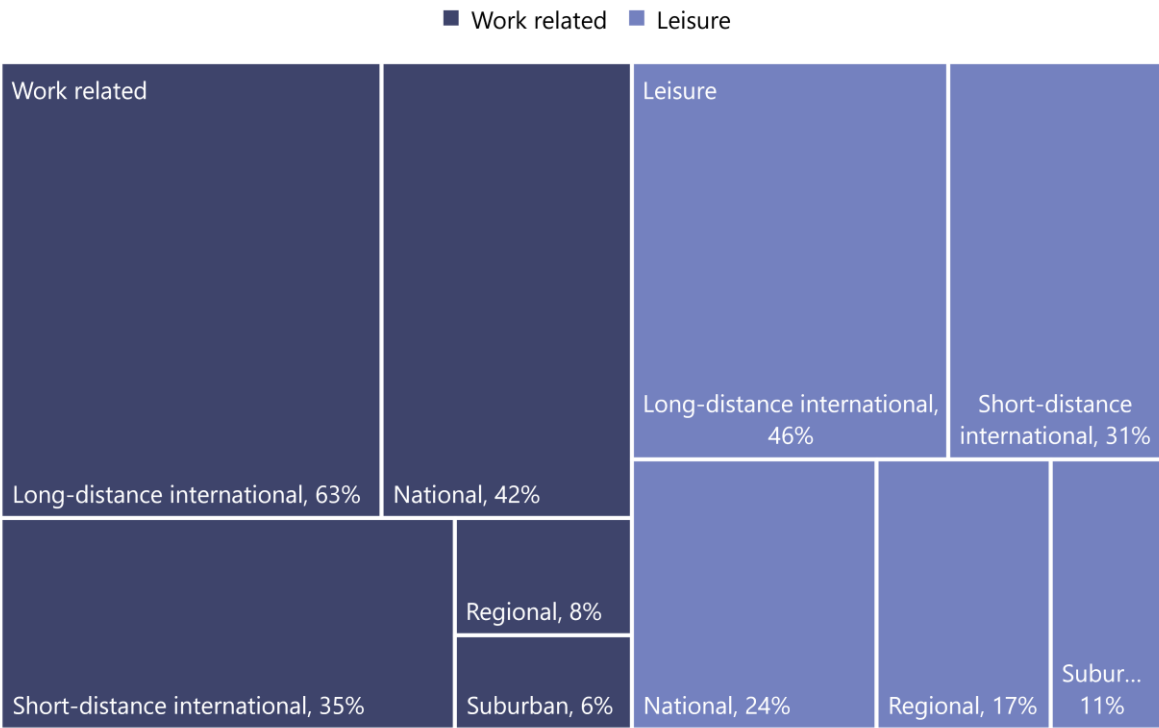


Figure 9.: Market share suitability per market segment by distance and trip purpose

Summarizing, the speed performance of hyperloop makes it most attractive for medium and long distances (regional, national and international trips). Medium distance market and long distance national and international markets should be considered as priority markets for hyperloop services. Careful consideration should be given to introducing hyperloop for short distance trips and for medium distance national trips with high-speed rail services available.

5.1.5 MARKET REQUIREMENTS

Service characteristic	Requirement
Throughput	20,000 passengers per hour per direction
Convenience	Frequent departures/arrivals, easy/few transfers, short egress/access time
Reliability / Punctuality	More reliable than rail and aviation.
Low transit time	Access times, trip time, transfer time.
Vehicle facilities	Include PRM, passenger restrains and lavatory access

5.2 PASSENGER SAFETY, REQUIREMENTS AND REGULATIONS

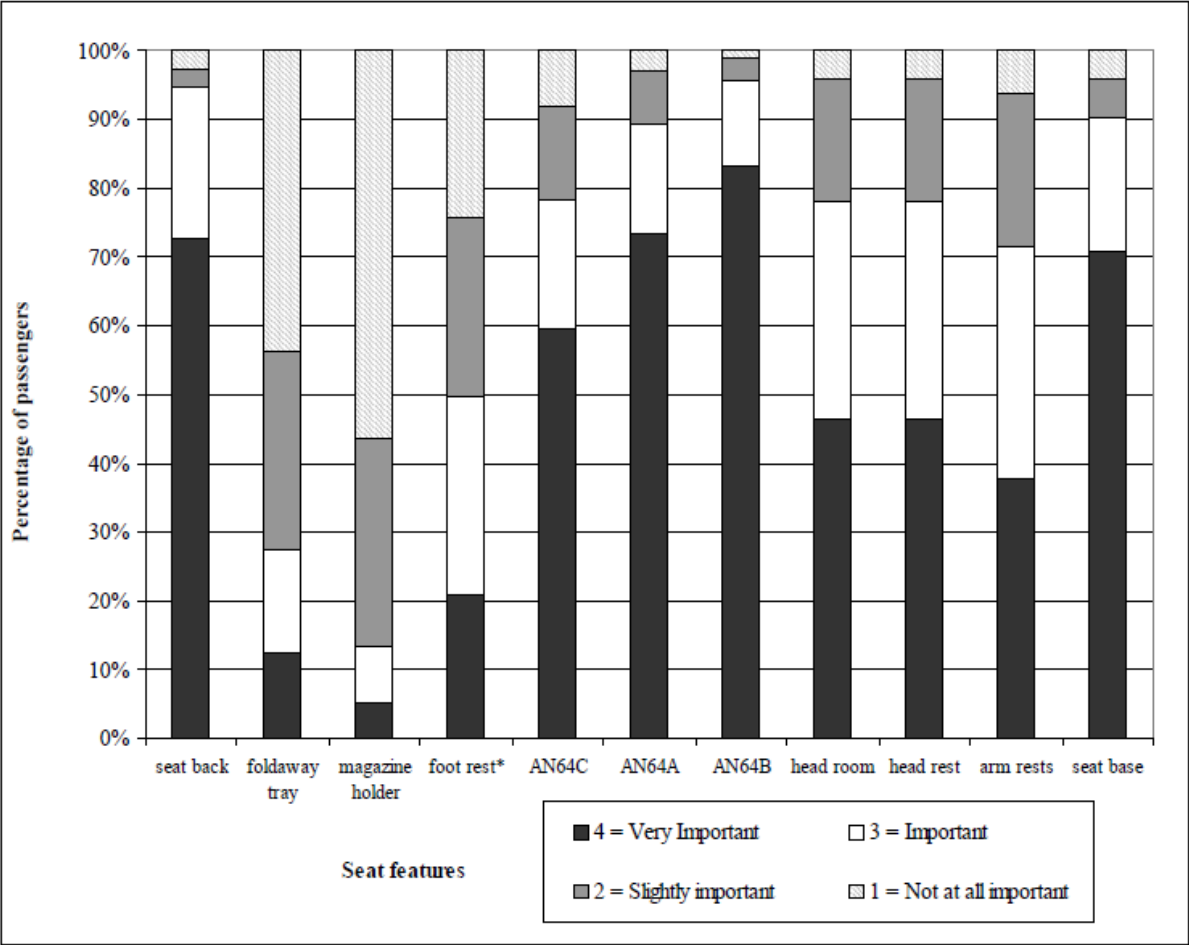
5.2.1 HUMAN FACTORS

The main objective of the vehicle interior design is to comfortably accommodate passengers and their belongings during a trip. High speed transportation systems have different solutions for this, mainly different seating configurations depending on ticket cost, distance, and type of vehicle making evident the different sensitivities.

Airlines usually try to cram as many people as they can in a single vehicle, while train cabins are more focused towards entry/exit since they have constant people flow. In the case of the hyperloop, the latter is more relevant due to hyperloop being less sensitive to weight than aviation.

The main trade-off is between passenger space (Related to comfort) and the number of passengers that can be fit into a vehicle (Passenger density). The latter is commonly more important to airlines than it is to train or bus operators. The seat design focus of the builders is to reduce weight and thickness of the seats themselves to accommodate more rows and reduce weight, all translating into higher profit for the airline operator.

To maintain the interior design at a "comfortable" level, there are regulations that dictate the minimum space that every seat should offer to the passenger. However, these regulations are old, and the population has grown in size, which makes them no longer adequate. (Quigley, Southall, Freer, Moody & Porter, 2007)



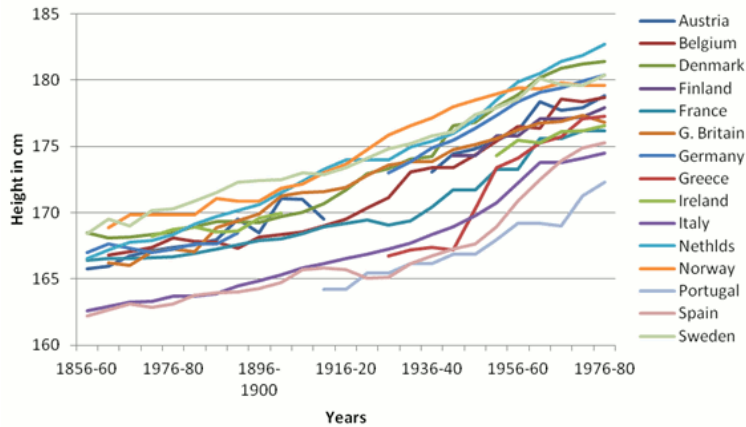
The features AN64A, AN64B and AN64C refer to the legroom, foot room and seat pitch, respectively. It is evident from the analysis that seat anthropometrics need to be revised to offer a comfortable passenger experience. Furthermore, the measurements should be in line with the prediction for population growth to future proof the system design.

<https://dspace.lboro.ac.uk/dspace-jspui/handle/2134/701>

Anthropometrics

From the introduction, it is evident that a right concept for anthropometrics must be taken to successfully design and develop a vehicle interior that meets the comfort requirements of the passengers. The basis for the layout distribution of the vehicle's interior, and consequently the exterior, is that of the P99th Dutch male and female.

The reason for this choice is that the population in the world, but especially in Europe continues to increase in stature given the European focus on healthcare research, resulting in a stature increase of 11cm in the last century (Hatton and Bray, 2010).



The trend is predicted to continue, which means that a P99 of today, will be a P95 in 50-100 years. It is not uncommon to assume that the hyperloop infrastructure will remain in service at least throughout the 21st century, in which case the design of the infrastructure must take into consideration the increase in population size to properly accommodate them in the future.

An overview of anthropometric measurements, relevant to a vehicle interior design is presented in this section.

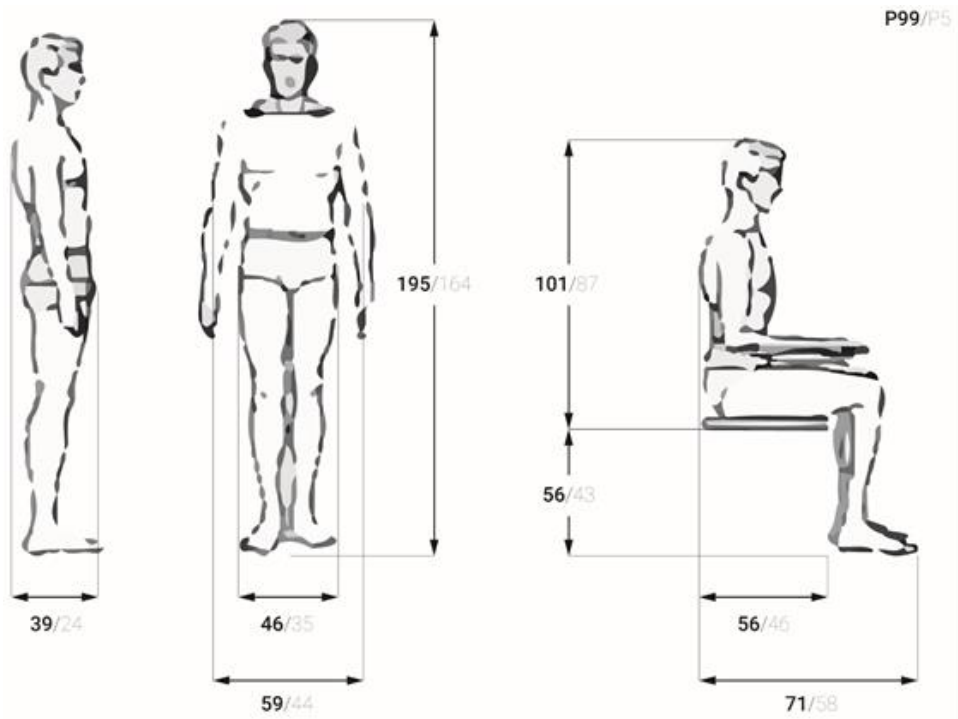


Figure 10. P99 Dutch Male (Measurements in CM)

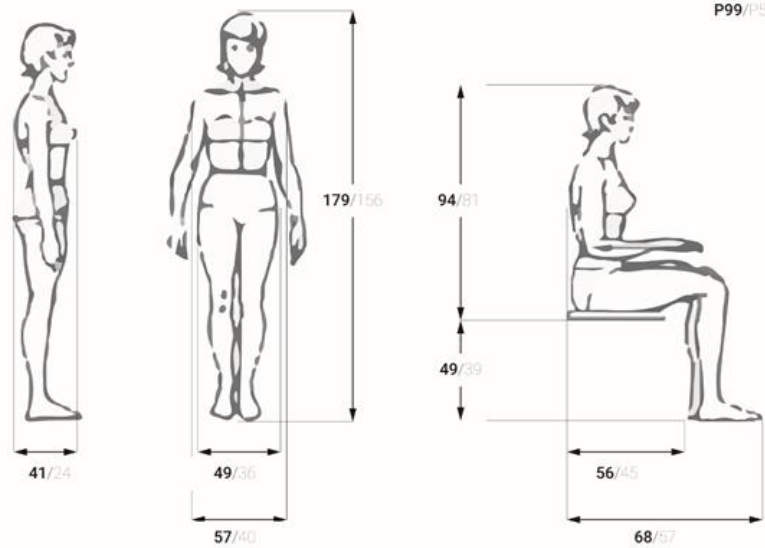


Figure 11. P99 Dutch Female (Measurements in CM)

These measurements will be used as a basis for the overall interior packaging concept. Depending on the relevance, P99/P5 Male/Female measurements will be used to develop a concept that is suitable for the highest amount of people possible.

5.2.2 REQUIREMENTS AND REGULATIONS

To construct a regulatory framework for the interior of the passenger hyperloop, a global survey of relevant regulations and standards is completed. An overview of this survey is given in the table below.

#	Regulation/standard	corresponding region/country	Classification
1	TSI PRM	EU	Contains several hard requirements which impact the LOPA and seat design.
2	TSI LOC&PAS	EU	Contains several hard requirements which impact the LOPA and seat design.
3	Elevators	EU	Not deemed relevant for passenger hyperloop interior due to multitude of fundamental differences.
4	Offshore structures		Not deemed relevant for passenger hyperloop interior but might be suitable for Hyperloop infrastructure.
5	Tunnel regulation	NL	Not deemed relevant for passenger hyperloop interior but might be suitable for Hyperloop infrastructure.
6	(High) speed trains like the Eurostar	EU	Covered by TSIs.
7	EASA CS-25	EU	Contains several hard requirements which impact the LOPA and seat design.
8	ASTM F2291	US	Regulations concerning amusement rides like rollercoasters can provide a broader/more general set of regulations if required for the design of the Hyperloop interior.

It should be noted that the scope is restricted to the interior the passenger hyperloop. If available, European or Dutch regulations are preferred due to the initial focus of hyperloop operations on this region of the world.

Based on this global survey, the following regulations and standards are evaluated in more detail:

1. EASA: Certification Specification (CS)

The European Union Aviation Safety Agency is an agency of the European Union with responsibility for civil aviation safety. Certification Specifications (CS) are non-binding technical standards adopted by the EASA to meet the essential requirements of the Basic Regulation.

2. Agency for Railways: Technical Specifications for Interoperability (TSIs)

The Technical Specifications for Interoperability (TSIs) define the technical and operational standards which must be met by each subsystem or part of subsystem in order to meet the essential requirements and ensure the interoperability of the railway system of the European Union.

3. Rollercoasters

The regulations and requirements posed on rollercoasters have been considered as well due to the following similarities with an Hyperloop:

- a. both transport passengers although rollercoasters commonly carry people from point A to point B instead of from point A to point B,
- b. rollercoasters are by definition one off product which is in line with the novel and innovative nature of the Hyperloop, and
- c. the (possible) conditions experienced by passengers, for example in terms of forces, are more alike than when compared with trains for example.

Lastly, the safety requirements for rollercoasters, while stringent, are more fundamental. This is due to the large variation between rollercoasters (one off products). They therefore rely less on already commonplace, technical solutions than the railway and airplane industry. Such standards can therefore create a safety vision and define the desired safety level for this method of transportation. This approach cannot be used when only aerospace regulations are considered since for some rules, like the 90 second evacuation requirement, no leading cause is known.

The regulations and standards for each of these industries identified as relevant for the interior of the passenger hyperloop are globally summarized in the table below. For a detailed overview in which these regulations are assigned to the requirements, critical points of attention and impactful design decisions identified in *ADSE HDP-WG-PAX-1.14*, the reader is referred to the corresponding Excel (*Overview regulations v1*).

#	Regulation/standard	Relevant sections
1	TSI PRM	4.2.2
2	TSI LOC&PAS	4.2.5, 4.2.10 - 4.2.12
7	EASA CS-25	3xx, 5xx - 8xx, 14xx
8	ASTM F2291	5, 6 - 8, 14

Based on this survey, no (critical) gaps in existing regulatory frameworks due to the innovative nature of the hyperloop could be identified for the interior of the passenger version. When requiring a more fundamental approach to safety, more exotic regulations like listed above for rollercoaster can be considered. Furthermore, when comparing this analysis to the *TR Draft Inventory & Map* sheet of the *TR Standards Inventory and Roadmap 2021-09-12* excel, only the regulations stated for rollercoasters are not present in the latter.

5.2.3 **EMERGENCY SCENARIOS**

The fundamental emergency scenarios for the passenger hyperloop are enumerated in the table below. The recommended emergency procedures for each of these scenarios are provided as well. It is assumed that the specifics of these procedures will comply with the applicable regulatory framework.

For this analysis, the evacuation of passengers via the tube to the outside world is deemed outside the scope of this working group and part of the infrastructure and operations. It's therefore assumed that, regardless of the environmental conditions in- and outside the tube, the safety and wellbeing of the passengers is harbored. Furthermore, for the evacuation of passengers via the tube, it is expected that they are able to exit the passenger hyperloop, and safely and easily move through the tube towards safe areas in a sufficiently small amount of time to harbor their safety in each of the situations identified below. The 'Technical Specification for Interoperability (TSI) SRT' and the 'Landelijke Tunnelstandaard' can provide guidance with respect to this aspect of the hyperloop design. It should be noted that these standards emphasize both prevention and mitigation to ease evacuation and rescue operations.

#	EMERGENCY SCENARIO	EMERGENCY PROCEDURE
1	On-board medical emergency	Depending on the type of medical emergency, a combination of the following procedures is applicable: 1. Provide appropriate on-board care with available medical equipment like first aid kits and an AED. 2. Navigate the passenger hyperloop to a location (easily) accessible by appropriate emergency services like the next station or perform an emergency stop. If required, pressurize the surroundings of the passenger hyperloop and have the emergency services either board the passenger hyperloop or evacuate the affected passenger(s) to provide appropriate care.
2	On-board fire	Depending on the specifics of the situation, a combination of the following procedures is applicable: 1. Extinguish the fire with available devices like a hand-held fire extinguisher, navigate the passenger hyperloop to the next station and evacuate the passenger hyperloop. 2. Pressurize the surroundings of the passenger hyperloop and evacuate the passenger hyperloop within the stipulated amount of time. NOTE: the effect of additional oxygen due to the pressurization of the passenger hyperloop should be taken into account.
3	Fire in the tube	This situation is not deemed likely due to the low-pressure environment in the tube and in itself not relevant for the interior of the passenger hyperloop. However, other emergency scenarios resulting due to fire in the tube might arise.
4	On-board loss of pressure	Depending on the specifics of the situation, a combination of the following procedures is applicable: 1. Provide oxygen to the passengers with, for example, an automatically deployed oxygen mask.

		2. Pressurize the surroundings of the passenger hyperloop and evacuate the passenger hyperloop. 3. Navigate the passenger hyperloop to the next station and evacuate the passenger hyperloop.
5	Loss of low-pressure environment in the tube	1. Perform an emergency stop and evacuate the passenger hyperloop (if further travel is impossible) or continue at reduced speed to the next station. 2. Provide (easy) access to appropriate emergency services.
6	Collision of multiple passenger hyperloops	1. Pressurize the surroundings of the passenger hyperloop and evacuate the passenger hyperloop. 2. Provide (easy) access to appropriate emergency services.
7	Derailment of a passenger hyperloop	1. Pressurize the surroundings of the passenger hyperloop and evacuate the passenger hyperloop. 2. Provide (easy) access to appropriate emergency services.
8	Explosion in a passenger hyperloop	1. Pressurize the surroundings of the passenger hyperloop and evacuate the passenger hyperloop within the stipulated amount of time. NOTE: the effect of additional oxygen due to the pressurization of the passenger hyperloop should be taken into account. 2. Provide (easy) access to appropriate emergency services.
9	Explosion in/near the tube	See #5: Loss of low-pressure environment in the tube.
10	Toxic gas release in a passenger hyperloop	Depending on the specifics of the situation, a combination of the following procedures is applicable: 1. Provide oxygen to the passengers with, for example, an automatically deployed oxygen mask. 2. Pressurize the surroundings of the passenger hyperloop and evacuate the passenger hyperloop. 3. Navigate the passenger hyperloop to the next station and evacuate the passenger hyperloop.
11	Loss of power	Depending on the specifics of the situation, a combination of the following procedures is applicable: 1. Pressurize the surroundings of the passenger hyperloop and evacuate the passenger hyperloop. 2. Navigate the passenger hyperloop to the next station and evacuate the passenger hyperloop.

5.2.4 **SAFETY CONCEPT**

The safety concept for passenger hyperloop is comprised of the following main points. These are explained in the following sections.

- The “brick wall” scenario.
- Pressure control
- Emergency evacuation
- Passenger restrains

5.2.4.1 **THE “BRICK WALL” SCENARIO**

If there is an accident, or an unknown situation in the tube, all the vehicles that are not involved should be able to come to a safe stop. This means, that to maintain safe passenger operation, the headway between passenger vehicles must be enough to allow vehicles to come to a full stop without reaching the point of the accident.

This is referred to as the “brick wall” scenario. In case of an emergency, accident, or an unforeseen circumstance, it is assumed that a “brick wall” has appeared on the track, so all vehicles must reach a full stop before reaching this wall. This is depicted in the following figure.

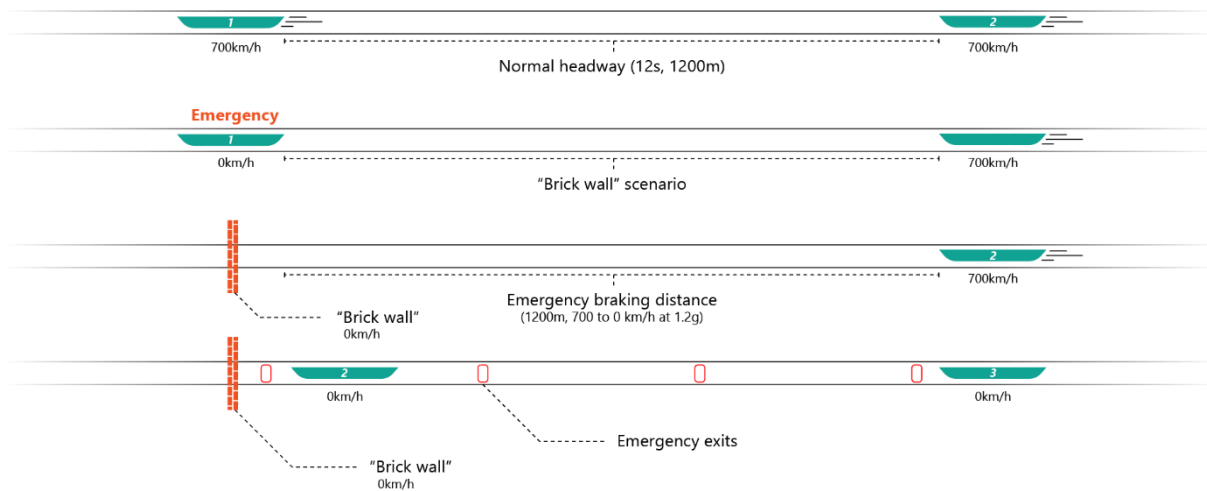


Figure 12. Brick wall scenario depiction

This approach ensures that emergency situations involve only one vehicle, and that there is always at least one infrastructure emergency exit available at each vehicle’s end. This is key to passenger safety.

5.2.4.2 PRESSURE CONTROL

The hyperloop vehicles travel inside a tube which during standard operation provides low pressure environment. This environment is close to 100Pa. This means that the cabin of the hyperloop vehicles needs to be pressurized, like an airplane.

However, an aircraft is constantly fed fresh air from the environment, and hyperloop vehicles do not have this option. This means that the latter need to have on-board oxygen supply as part of the environmental control system.

Furthermore, in case of an emergency, the following measures are to be taken, fully or partially, depending on the case.

- Provide emergency oxygen to passengers
- Bring the tube environment to an ambient pressure

5.2.4.2.1 EMERGENCY OXYGEN SUPPLY

Oxygen is fed to the vehicle cabin by means of a reservoir, part the environmental control system. Amongst other things, this system is responsible for the filtering and supply of air inside the cabin. Emergency oxygen access must be provided to passengers in case of an emergency.

5.2.4.2.2 RAPID TUBE PRESSURIZATION

Depending on the emergency case, the repressurization of the tube may be necessary in order to allow passengers to evacuate. This is done by means of pressure valves, these valves will open in case of an emergency and air will be allowed into the infrastructure, flooding the tube.

The pressure valves are located throughout the entire length of the infrastructure. The exact specifications are still unknown. A diagram of a 24m long section of tube is depicted below.

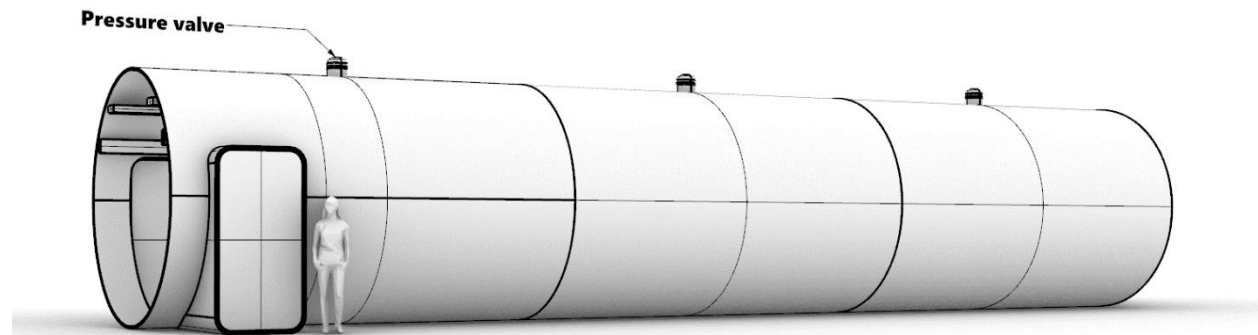


Figure 13. Tube overview, showing pressure valves and emergency exit.

This process will only occur if the passengers are required to evacuate the system during an emergency. The tube infrastructure will be segmented to allow for repressurization of specific segments in case of an emergency.

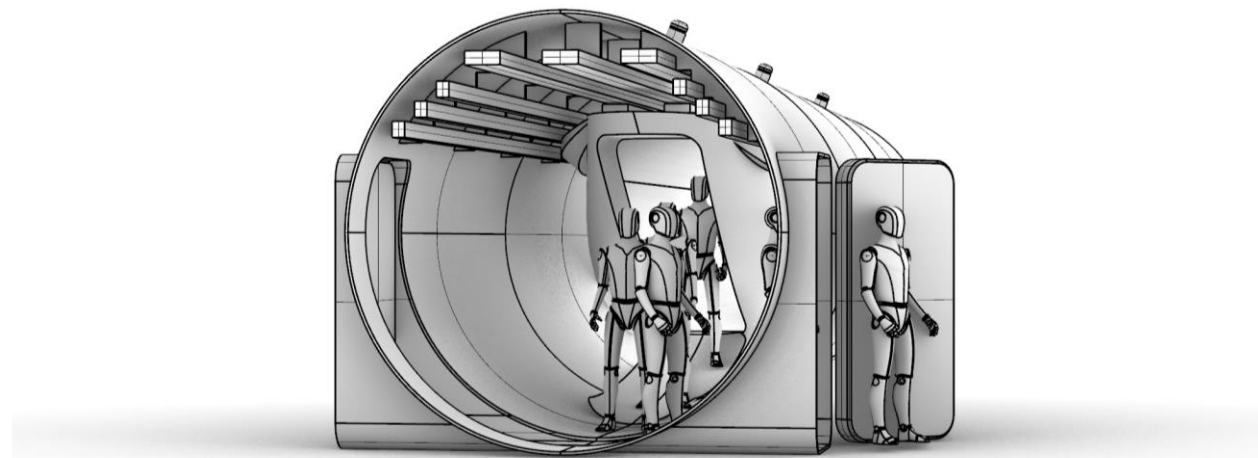


Figure 14. Passengers exiting the vehicle and infrastructure during emergency.

5.2.4.3 EVACUATION

Infrastructure

Emergency exits need to exist at regular intervals throughout the entire infrastructure. These exits need to provide safe passage from the interior to the exterior of the tube at a ground level. The distance between the emergency exits is dictated by the distance between the vehicles, the emergency braking rate and physical evacuation requirements.

Vehicle

All vehicles must have at least two emergency exits that allow passengers to evacuate to the interior of the tube. These exits are to be located at the nose and tail of the vehicle and will allow passengers to exit either or both directions depending on the scenario.

The spacing between the vehicles, given by the headway and the emergency braking will ensure that there is always an infrastructure emergency exit safely reachable by the vehicle front. As a redundancy, another exit must always be available via the vehicle rear, as depicted in the following image.

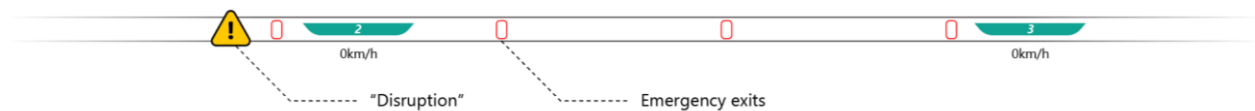


Figure 15. Emergency evacuation scenario

5.2.4.4 RESTRAINS

Passengers traveling with the hyperloop will be required to wear a seatbelt at all times. The enforcement of this requirement is still a point of discussion; however, its need is clear when considering the speeds at which the hyperloop is going to travel, and the braking rates that it can achieve in case of an emergency.

It is likely that a stand similar to aviation is taken, in which passengers are allowed to stand and walk around during cruise and need to strap in during acceleration, braking and switching. These details, however, require further analysis.

Furthermore, similar to aviation, all hand luggage must be restrained to remain in its place during an emergency event.

5.2.4.5 OTHER SAFETY CONSIDERATIONS

5.2.4.5.1 PRM SAFETY

PRM- Person with Reduced Mobility

While general access and accommodation is a factor the required safety measures have the biggest impact on the Lay-out Of Passenger Accommodation.

Considering the survey of relevant regulations and standards (chapter 6.2.1), the Technical Specifications for Interoperability (TSI) PRM is identified as the only stand-alone reference dealing with PRM, i.e., for other sources PRM are not considered explicitly but taken into account while creating the overall guidelines and regulations. Particularly section 4.2.2 provides appropriate guidelines for the interior and LOPA of the passenger hyperloop for dealing with PRM (safety). These specifications cover topics ranging from the LOPA to the visual indicators and are highlighted in the ADSE Summary TSI LOPA_20211216 presentation. For a detailed overview in which these guidelines are assigned to the requirements identified in ADSE HDP-WG-PAX-1.14, the reader is referred to the corresponding Excel (Overview regulations v1).

During a talk with an ADSE colleague who is in a wheelchair, autonomy, especially in emergencies, is furthermore identified as a key aspect. While emergency procedures are provided to air travelers, this is not the case for passengers using a train. In both cases, the PRM is dependent on help from other travelers or personnel although this is not described in a protocol for the latter.

5.2.4.5.2 PRM SEATING

Here we have two options. The regulations for trains allow passengers to travel in their own wheelchair strapped but not fixed into place, because forces are low. In aviation, every passenger must sit in a crashworthy seat, therefore cannot sit in their own wheelchair, even if there was space provided. Hence the aisle chair, which is merely to transfer a PRM from the jet bridge through the aircraft to a standard, crashworthy seat (and to the lavatory of course). There are concepts of crashworthy wheelchairs, including secure mounting (a bit like ISOFIX). However, these concepts never made it to an actual certified solution.

Evacuation

Depending on seating scenario, the emergency exit routes would either:

1. PRM sitting in regular seat (via aisle chair) -> emergency exits can be normal (narrow) too.
2. PRM sitting in standard or special wheelchair -> now the emergency route can or has to be bigger, depending on how the PRM is supposed to be evacuated. This scenario would assume there is no aisle chair on-board, so transferring via a narrow aisle is not possible.

It is recommended to use the latter solution. It is expected to maintain embarking and disembarking times low and provides autonomy to a PRM during an evacuation procedure.

5.2.4.5.3 SAFETY EQUIPMENT

A list of possible emergency scenarios and their suggested emergency procedures was provided. SDC in turn has then analyzed this list and determined the consequences for the cabin and LOPA design.

Several items can be added to the cabin to increase safety in emergency scenarios, both in accident avoidance and in mitigation. These include fire extinguishers, smoke detectors and fire suppression systems, axes to clear debris, as well as preparation for non-system-related emergencies, e.g., first aid kits and defibrillators.

A key element in safety is communication, especially when no trained staff is on-board. Passengers need to be well informed on what emergency or disruption has occurred, what measures are being taken from the operator, and which eventual measures the passengers have to perform themselves (e.g., wearing an

oxygen mask or opening an emergency exit). This requires a general safety briefing to prepare for, and precise assistance in case an emergency actually occurs.

Emergency exits and escape routes must be well marked, lit and powered by a redundant power supply. Indicators should also inform about cabin and tube air pressure and quality in order to guide the passengers through an emergency scenario.

5.2.4.5.4 SAFETY ENFORCEMENT

As listed above, it can be expected that the safety measures required are stricter than in railways, mainly due to the pressurized hull design and the more dynamic deceleration, resulting in higher forces during emergency stops. Therefore, safety measures can in many ways expected to be more similar to those applied in the aviation industry.

Unlike in commuter trains or subways however, in aviation, a crew is always present to ensure on-board safety, not only by reacting to emergency scenarios and providing assistance in case emergency procedures such as an evacuation need to be executed, but also by strictly enforcing all the built-in safety measures in the first place. The importance of staff in terms of safety is discussed in a segment below.

In case cabin crew is not present in the hyperloop vehicles, a highly likely scenario especially for smaller vehicle concepts, it needs to be considered how mandatory safety measures, like wearing a seatbelt or storing all luggage safely, can be monitored or enforced. If a safety measure is considered so crucial that the vehicle cannot start, the non-compliant passengers must be addressed without causing delays which would endanger a smooth and efficient operation of the entire system.

In aviation, cabin staff is always present to inform, to prepare and to aide passengers in case of an emergency, e.g., loss of cabin pressure or a rapid evacuation.

Again, comparing to aviation, passenger aircraft are mandated to be evacuated within 90 seconds after the call for evacuation has been made by the captain. Well-trained flight attendants are in charge of opening the emergency exits, which trigger automatic pneumatic slides below. They then aide passengers in departing the aircraft, especially passengers with reduced mobility (PRM).

The 90 second rule is valid for an evacuation with only one side of the aircraft's emergency exits open, due to emergency scenarios such as a fire in one of the engines or wing fuel tanks.

From what's discussed above, it is obvious that in aviation, staff plays a vital role in calling emergencies, decision making and guiding passengers to safety. Therefore, with Hyperloop's emergency scenarios in many ways similar and comparable to those in aviation, the presence of staff from a safety aspect, among other benefits, should be considered. With emergency exits more limited than in aircraft cabins of comparable size, the role of trained staff conducting an orderly evacuation can be even more significant and potentially lifesaving.

5.2.4.5.5 PERCEIVED SAFETY

Although in the best-case scenario passengers would not require following any safety procedures as accidents are extremely rare, it can be important to communicate the safety measures taken in order to gain the passengers trust and confidence into the systems technology.

Since Hyperloop is a fairly new and high-tech concept operating at high speeds and within a hostile low-pressure environment, some passengers or even potential customers might be interested in what safety measures are actually in place for different emergency scenarios. This can range from explanatory videos to safety briefings such as those mandatory on all commercial flights, as well as simply informing interested passengers about the vehicle's current status such as speed, acceleration, oxygen supply level, cabin air



quality, ETA, and so on. This can be of additional importance due to the fact that once inside the vehicle, all reference and orientation towards the outside world is lost, as the vehicle or the tube, by design, does not provide any windows.

5.3 PRODUCT DESIGN AND PASSENGER EXPERIENCE

5.3.1 CRUCIAL FACTORS

During the process described in this report, especially during the phase of LOPA development, a number of factors began to stand out which appear to be significantly more impactful to the design than others. Below is a list of those crucial factors described in more detail.

5.3.1.1 TURNAROUND TIME

To be able to meet the required throughput, turnaround time must be short. Based on rail current practices and the expected technical capabilities of hyperloop, stop over time is 3 minutes.

5.3.1.2 LUGGAGE

Personal items and small carryon pieces of luggage can be carried onboard. Large and odd sized luggage should be sent via a different service, outside of the scope of this document. See chapter 5.3.4.1

5.3.1.3 STAFF

The presence of staff has a decisive impact on a variety of topics: passenger safety and safety enforcement. While a significant cost factor, staff can also be a significant factor in providing a smoothly running system. Two options were considered:

- **Platform staff.** Platform staff remain at the station and do not travel with the vehicles, similar to staff at theme park rides. They can facilitate ingress/egress for PRM, help finding seats and luggage storage and generally motivate passengers for a timely process.
- **On-board staff.** On-board staff travels with a vehicle, similar to flight attendants in the aviation industry. They can provide all the services of platform staff, but also provide assistance during the journey, especially during emergency situations such as evacuations.

The option of only using platform staff is favored due to costs.

5.3.1.4 INGRESS/EGRESS SIMULATION

With ingress/egress times being such a crucial factor, a simulation or real-world test should be conducted. It could help finding values for variables such as:

- general realistic ingress/egress time
- passenger discipline (willingness to not delay departure)
- required door count
- longer aisles, more aisle/single seats vs. shorter aisles, less single seats.
- luggage storage
- factor of delay for PRM
- impact of no staff / platform staff / on-board staff

The aviation industry for example has conducted several studies on improving boarding time. One of the software used for simulations is PAXelerate from German research group 'Bauhaus Luftfahrt', which could potentially be modified to be used in the context of Hyperloop.

[PAXelerate | Boarding Simulation by Bauhaus Luftfahrt e.V.](#)

5.3.2 S. M. L SCENARIOS

In order to assess the possibilities of a passenger hyperloop system, a set of scenarios was created. The main difference between them is the overall dimensions of the vehicle, consequently, the dimensions of the infrastructure. An XL version could have been developed also, with the purpose of transporting containers, however, insufficient argumentation was found in its favor, therefore, an XL scenario was not developed.

The most important constrain is the cross-sectional area, since this defines the limits of capacity in two of the three dimensions. Vehicles may be shorter or longer in the future (Up to a certain limit, unknown at the moment) however if the cross section is set, it will be very difficult to modify later on. Hyperloop is specifically sensitive when it comes to this metric, unlike other modes of transport which have enjoyed more freedom, albeit with sensitivities of their own, such as aviation and rail.



Figure 16. overview of hyperloop concepts, L (Top), M (Middle), S (Bottom, 3 vehicles)

The dimensions set for the different scenarios are mainly based upon and correlate in width with the number of seats per row (1+1, 1+2 and 2+2) and in terms of length with the desired total passenger capacity per vehicle. This number and the resulting physical dimensions are linked directly to different usage scenarios and passenger experiences, such as, for example, operating intervals to a certain destination.

During this process, the goal was to keep the three scenarios as far apart from each other as possible, in order to keep them distinguishable from each other and recognize clear differences, advantages and disadvantages. However, a conclusion or synthesis of those three variations may result in a compromise between the scenarios listed above, trading features and solutions of different vehicles to achieve an optimal result.

These scenarios were compared against rail and aviation to understand what the inherent place of hyperloop could be in the mobility landscape. Afterwards, the different hyperloop concepts were compared to each other and against a perfect benchmark, dubbed “paradiseloop”.

5.3.2.1 PARADISELOOP

- Beautiful
- Perfect
- zero emission during construction
- Zero emission during operation
- Unlimited capacity
- Free to build and use
- Godlike

5.3.2.2 HYPERLOOP S

Outside the more homogeneous group of M, L and the previous HARDT concept, Hyperloop S stands out with its overall small dimensions and resulting cabin space. This is due to the fact that Hyperloop S is supposed to find the lower feasible boundary of passenger and cargo transport. All learnings from studying and exploring Hyperloop S can be used to evaluate the feasibility of running a passenger service in a 2.5m tube.



Figure 17. Hyperloop S overview (3 vehicle platoon)

The limited diameter creates several disadvantages compared to its bigger siblings M and L. Two seats per row are possible, yet the aisle width is limited, and due to the small tube diameter, a significantly curved cabin sidewall interferes with the passenger’s leg and headroom. Styling measures like light materials and ample cabin lighting have to be taken to combat claustrophobia among passengers and making the cabin to appear bigger than it actually is.

Still the cabin is roughly comparable to a small business jet, an icon of luxurious and desirable mode of transport. Small business jets can also be referenced for lavatory design. While limited in many factors, a fold-away lavatory can be installed, in which a door closes off part of the aisle providing the necessary space and privacy.

Hyperloop S concepts have a transport capacity of around 8 passengers depending on the LOPA.

5.3.2.3 HYPERLOOP M

Placed between Hyperloop S and L sits Hyperloop M with a tube diameter of 3.4 m and a resulting cabin width of around 2.5 m. This places this setup in a category similar to coaches, and a seating layout of 2 + 1 is appropriate. While significantly more spacious than Hyperloop S, aisle height is limited, further amplified by the fact that due to the uneven seating layout, the aisle has to be placed slightly off-center from the cabin's ideal maximum height. Hyperloop M allows basic handling of PRM passengers in

wheelchairs; however, this would limit sections of the cabin to a 1 + 1 layout. Due to the curved sidewall, leg- and shoulder room is slightly more limited than in square section coaches. Regular, fixed-room lavatories are possible, although making these PRM accessible will be a challenge.



Figure 18. Hyperloop M overview

Hyperloop M concepts have a transport capacity of around 40 passengers, depending on the LOPA.

5.3.2.4 HYPERLOOP L

With a tube diameter of approximately 4.3 m, Hyperloop L is placed right at the top end of HARDTs envelope of feasible tube diameters. The resulting cabin size is similar to a modern commuter jet and features a 2 + 2 seating layout. While still smaller than a common Boeing 737 or Airbus A320 cabin, it provides the best perception of space and as well as the best aisle height for tall passengers among all three Hyperloop concepts. A circular cross-section as used on many airliners is not necessary, which helps with leg and shoulder room compared to the aviation examples provided.

However, it is still significantly smaller than a typical train cabin, which is not only wider, but features an almost square cross section, as restrictions due to pressurized hull design do not apply. Due to its larger size, Hyperloop L also provides the best setup for accommodating PRMs including those in wheelchairs. Given the width between the aisle and the cabin side wall, a more generous lavatory can be installed, better suitable for PRM and probably even for wheelchair access.



Figure 19. Hyperloop L overview

Hyperloop L concepts have a transport capacity of around 60 passengers, depending on the LOPA

HARDTs previous concept is, in most factors, placed between Hyperloop M and L. With Hyperloop S being the special case at the minimum tube diameter, M and L are designed to show the opportunities of either decreasing or increasing the size of the previous HARDT concept. Layout of passenger accommodation

The Layout of Passenger Accommodations (LOPA) is a floorplan of the cabin with all components such as seats, doors, lavatories, luggage racks and safety equipment such as fire extinguishers and emergency exit included.

The LOPA is influenced by various factors such as vehicle size, passenger capacity, desired passenger comfort, service levels (classes), vehicle equipment, vehicle operation, staff presence, passenger flow, emergency scenarios, PRM accommodation requirements and more.

Early in the process, it became obvious that all the variables related to the topics above were interconnected and influenced the three different layouts S, M and L in different ways and magnitudes. In order to break this chicken vs. egg situation, the decision was made to develop three preliminary LOPA design sketches

These loose and open sketches could then be used as a base for discussion across different topics, start conversations, highlight issues and challenges, and eventually trigger ideas and finding solutions. Therefore, for this project, a method of a modular LOPA design was used for prototyping: Build-a-LOPA.

Different interior items were grouped, and general dimensions researched. The items were then placed in segments of the three projected cabin diameter sizes S, M and L. Then, several of those segments were lined up and combined to a LOPA. The elements were interchangeable within their size group and allowed for different arrangements of e.g., seats (mainly in groups of 12), lavatories, special PRM seats including companion seats, door areas, and so on. While meeting loose predefined passenger capacity targets, the various combinations led to different vehicle lengths within their width category and were able to be tested against cabin regulations and general usability questions.

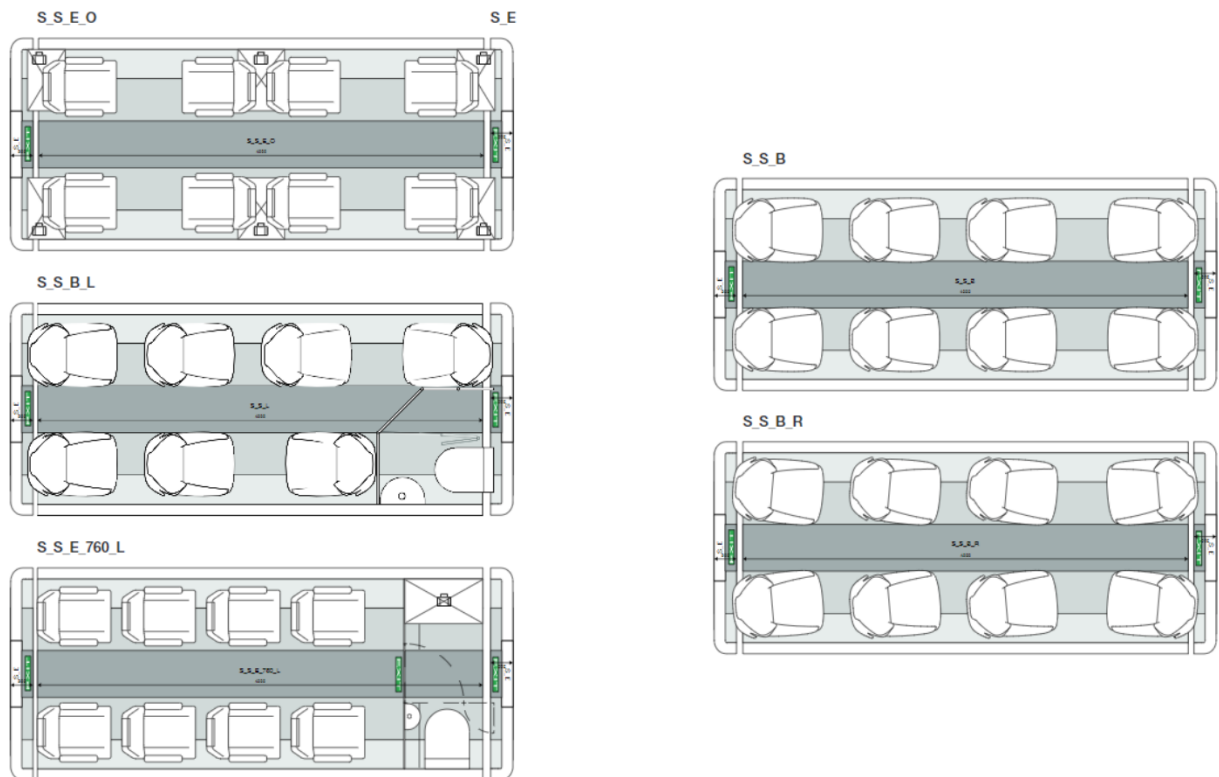


Figure 20. Hyperloop S LOPA examples

In these examples it can be observed that the end sections are similar in all the layouts, however, the middle sections can be different, showing different seat designs or even a lavatory and luggage section.

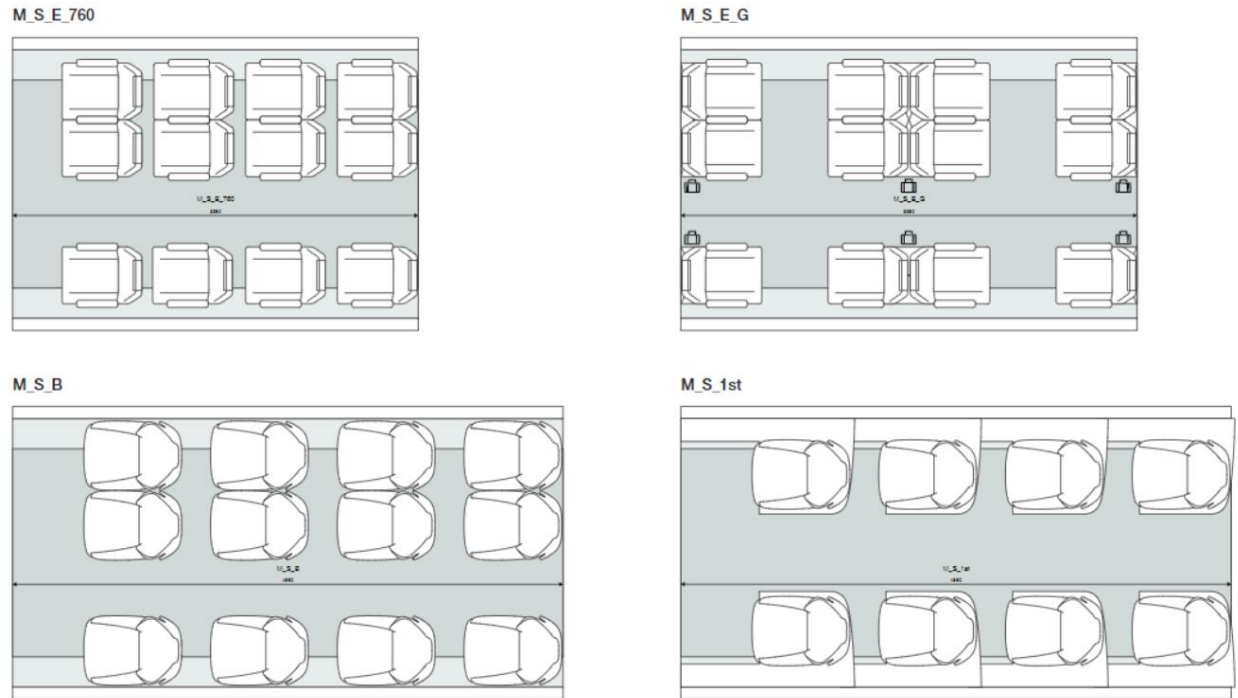


Figure 21. Hyperloop M segment examples

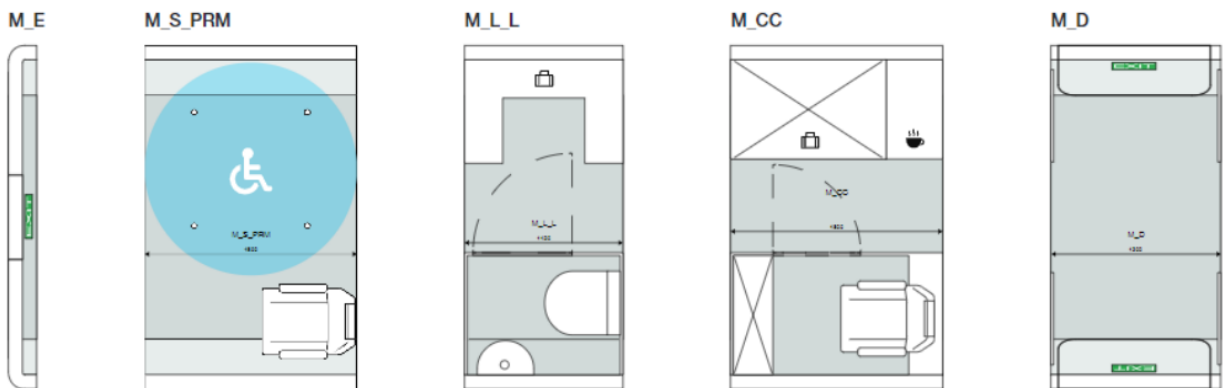


Figure 22. Hyperloop M sections. End, PRM, Lavatories & Luggage, Cabin Crew and Door

All of the M LOPAs and components can be interchangeable to produce and end vehicle layout. With the M concept, there is more variation, as the doors, lavatories, luggage, PRM sections and others can be moved around and swapped for one another.

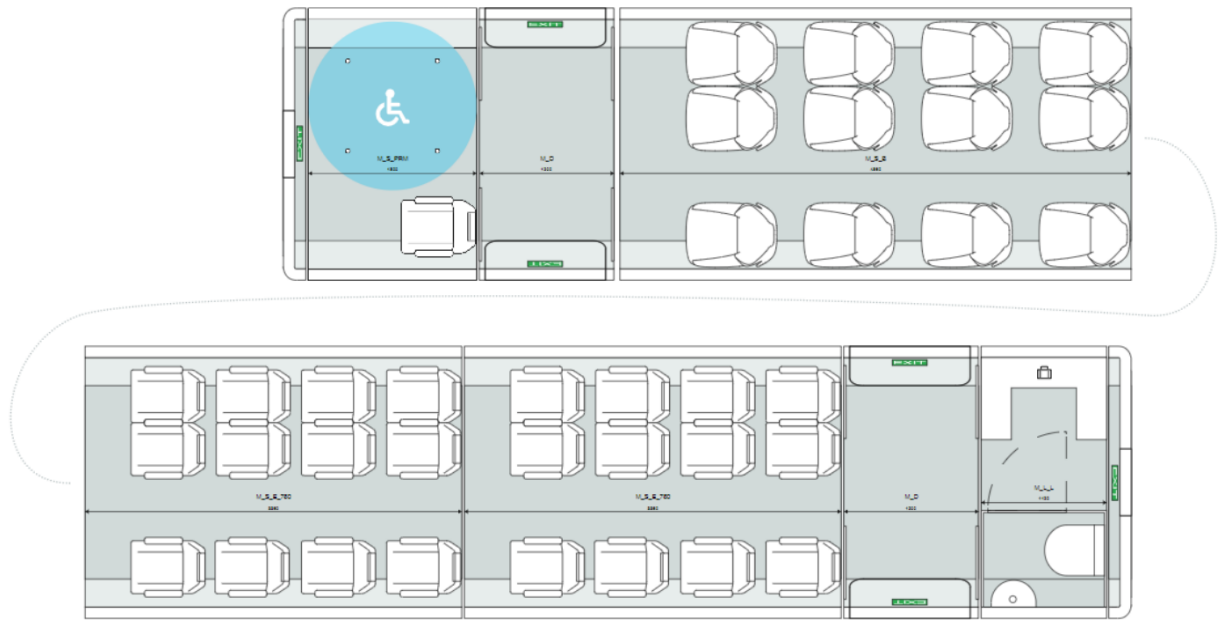


Figure 23 Hyperloop M LOPA example

The figure above shows an example of a LOPA belonging to an Hyperloop M concept. The segments have been matched to provide accommodation for 37 passengers and 1 passenger with reduced mobility. Two doors can be seen close to the ends of the vehicle as well as a lavatory and luggage module.

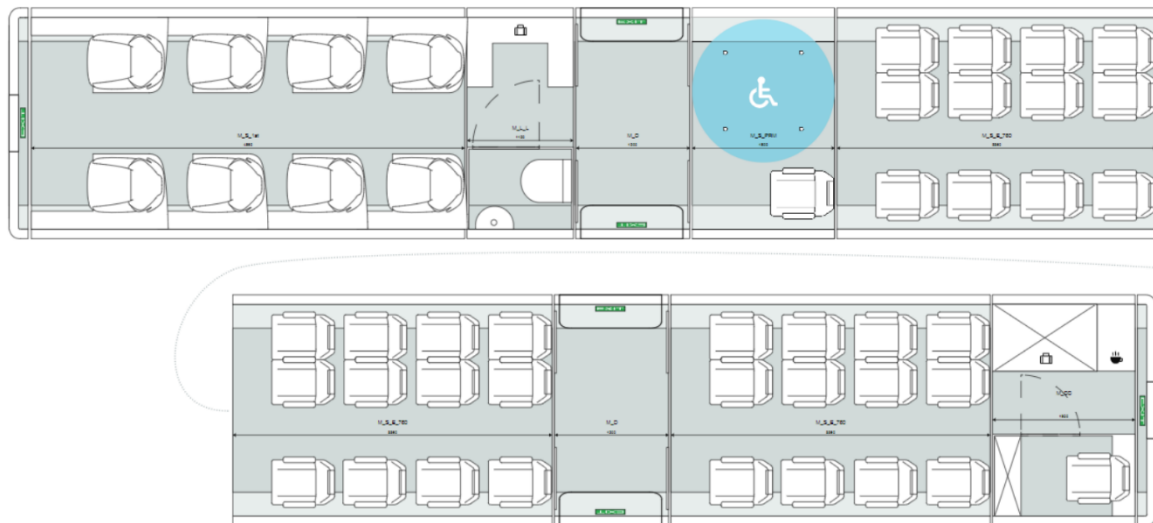


Figure 24. Hyperloop M LOPA example 2

The figure above shows an M vehicle with higher capacity, 46 passengers, 1 PRM and 1 crew member. Below, an even larger LOPA, belonging to hyperloop L can be seen.



Figure 25. Hyperloop L LOPA example

The image above shows a LOPA for Hyperloop L. 62 passengers, 1 PRM, 2 cabin crew. These layouts were compared against each other during the trade-off process.

As a preliminary result, the findings of those tests against the regulations (e.g., TSI) were summarized and evaluated. Some regulations were passed with the proposals provided, some were considered in a margin where a slight and feasible update of the LOPA or one of its segments would make it pass the requirement, while in some cases, the concept by design could under no circumstances meet the regulations.

Furthermore, different classes could combine various seats and seating arrangements in a single vehicle. Apart from other features like different materials and premium services provided, different classes would provide an opportunity to widen target groups. Classes can be considered from an economical point of view, from a corporate identity aspect, meaning whether the operator would like to provide different services and experiences to higher paying customers or include all passengers into the same offer. From a vehicle architecture and LOPA point of view, classes can help in balancing vehicle length.

5.3.2.5 TUBE & VEHICLE DIAMETER

The tube diameter is a crucial variable in the system design. It impacts the vehicle and therefore cabin width and height, but also the entire networks build and operations cost. A wider tube offers more passenger comfort and allows for more flexibility in seating arrangements. It also facilitates accommodation of PRM (people with reduced mobility).

On the flipside, a wider tube diameter heavily increases the cost per km of the tube. It also increases operation cost, as more air volume is required to be evacuated from the tube across the entire network.

From a passenger point of view, a bigger tube diameter is preferable. From an operator's point of view, a smaller tube is preferable.



Figure 26. S, M, L Tube diameters

5.3.3 CARGO HYPERLOOP INTERFACE

The HDP envisions a hyperloop capable of transporting cargo and passengers in the same infrastructure. Although vehicles for cargo and passengers are expected to be different, the requirements from both have to be considered for the design of the infrastructure.

5.3.3.1 PALLETIZED CARGO REQUIREMENTS

According to the “Market needs for cargo hyperloop capabilities” report, 92% of palletized cargo is packaged under a total height of 2 meters, including the height of the pallet. Also, 90% of the cargo is packaged under a total width of 1.2 meters. Please consult Appendix B for more information.

The minimum requirement for a hyperloop cargo envelope is 1.2 meters wide, by 2.0 meters high, plus tolerances. Length is not considered for this tradeoff.

5.3.3.2 AIRCRAFT CARGO REQUIREMENTS

Aviation has a different format for cargo transport, defined as ULD, Unit Load device. These devices come in a range of different sizes. At this point we have no data regarding the usage distribution of the different types of ULDs, in other words, we do not know which models of ULDs are used more often in the industry. We assume, from conversations with different people that the ULD3 is the most used. More research is needed in this area.

5.3.3.3 CROSS SECTIONS

To understand how the passengers and cargo would fit in each of the concepts, some cross section drawings were made and are presented below. These drawings contain the high-level parameters that are

given by the engineering team, including high-level packaging, the vehicle motion envelope and the infrastructure tolerance. The latter two are marked with dotted lines.

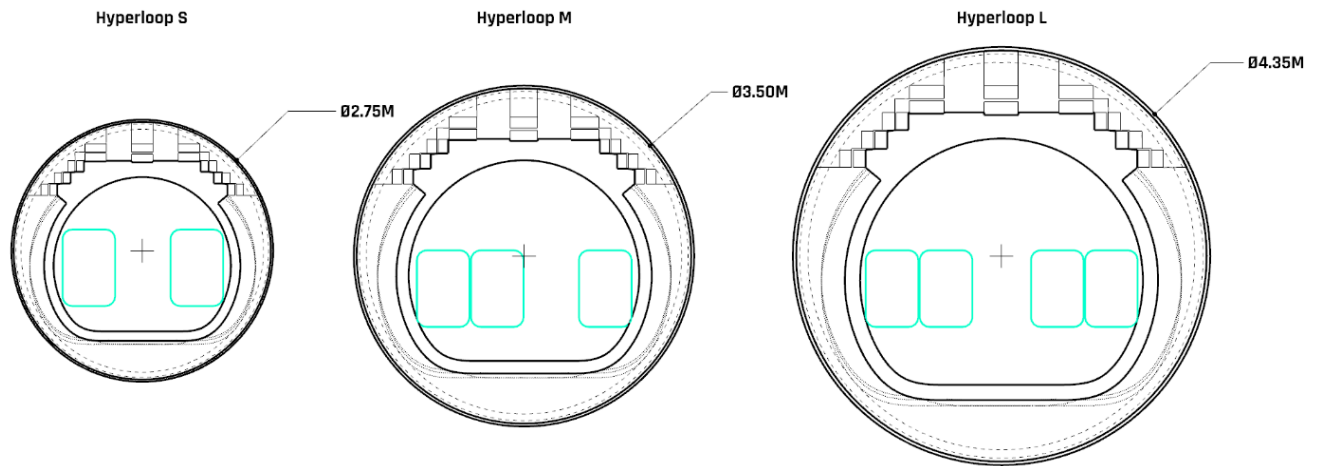


Figure 27. Hyperloop concepts cross sections with seats

The cross sections are based on passenger transport, however, the hyperloop will also transport cargo in the end. The ability of each of the concepts to transport cargo (In separate vehicles) could make one of the concepts more suitable for development.

In the following cross section drawings, the concepts are depicted in combination with the cargo cross section. This cargo cross section corresponds to the requirements set in the previous sections and it is aligned to best fit the cross section of the vehicle.

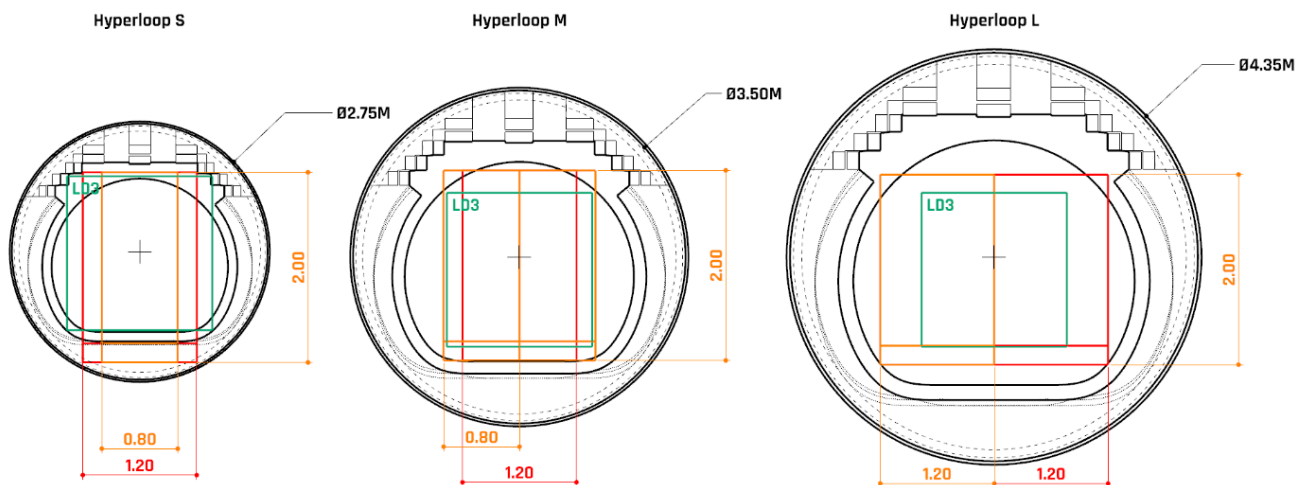


Figure 28 Hyperloop concepts cross sections with cargo footprint

The image above illustrates how would cargo fit in the given concept envelope. It is evident that Hyperloop S concept is unable to meet the cargo requirements for palletized transport or ULD LD3. In order for Hyperloop S to meet cargo requirements, an updated cross section has been drawn and is presented below.

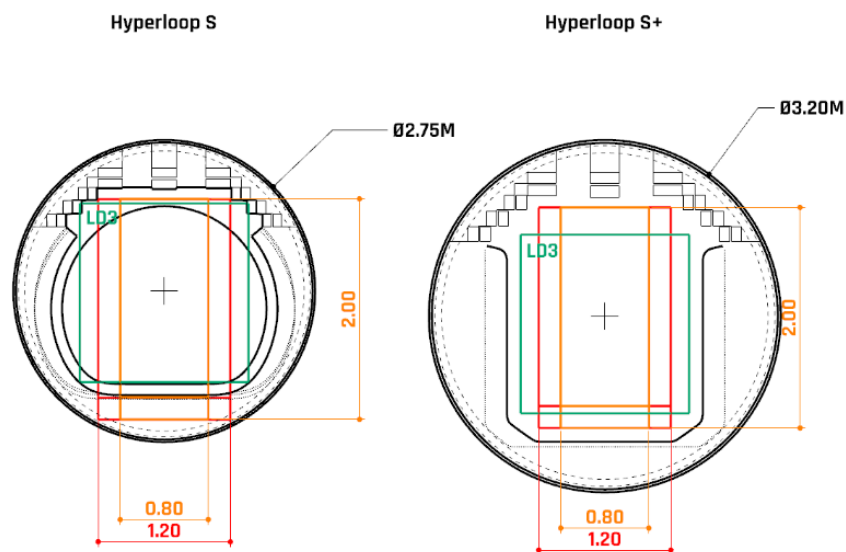


Figure 29 Hyperloop S and S+ concepts cross sections side by side

The tube infrastructure would have to grow to 3.20M in diameter to be able to accommodate the requirements for cargo transport. The tube's diameter of this concept, dubbed Hyperloop S+, would be only 20cm smaller than Hyperloop M. All updated cross sections are presented below for comparison.

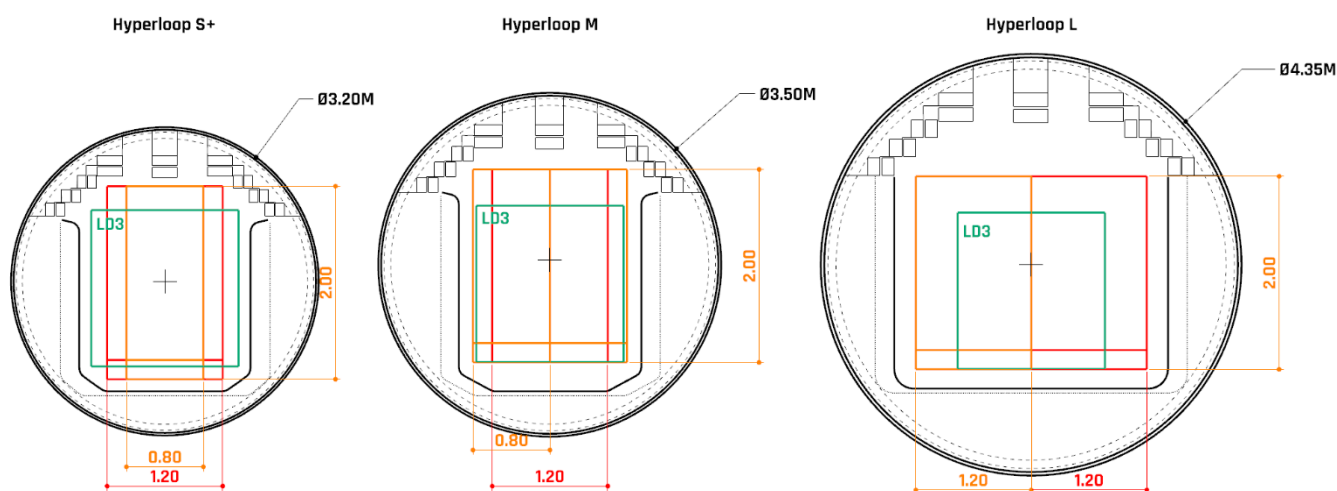


Figure 30. Updated hyperloop concepts cross sections

With the updated cross sections, all S, M, and L concepts are able to meet the requirements of cargo transport. Also, they would be able to meet the requirements of passenger transport, however, hyperloop S would be wasteful in terms of passenger transport, being too large for 2 passengers and not large enough for 3. It is notable that both Hyperloop M and L are able to transport 2 cargo pallets side by side.

5.3.3.4 SAFETY (PASSENGERS AND CARGO)

To maintain the safety level of the system, the safety concept for passenger transport remains the way it is. This means that a cargo vehicle or platoon must always respect the headway defined by the passenger hyperloop safety concept.

5.3.3.5 CONCLUSIONS

When considering cargo transport, there were two main requirements set for the hyperloop concept cross sections.

- To fit palletized cargo up to 2m in height
- To fit LD3 ULD

Only Hyperloop M and Hyperloop L meet these two requirements. Hyperloop S is required to grow from 2.75m in tube diameter to 3.2m in order to fit these requirements in Hyperloop S, leaving it only 30cm smaller than Hyperloop M. Hyperloop M and L are both capable of transporting 2 euro pallets side by side, although L can accommodate them transversely, and both concepts are able to transport LD3 ULDs, furthermore, M can transport up to LD29 and L up to LD39 (One size bigger than LD29, and the absolute biggest.)

Given the 85cm jump from M to L, the given requirements, and the marginal improvement in cargo transport capacity, hyperloop M is the best solution for cargo transport. An argument could be made for the Hyperloop L if the throughput of Hyperloop M is not enough. Hyperloop S is not recommended.

5.3.4 OTHER CONSIDERATIONS

5.3.4.1 LUGGAGE

Depending on the seating arrangement, luggage can be stored under the seats, between seats (back-to-back arrangement) or in luggage racks placed at the end of each section.

To keep the stop-over time as short as possible, passengers must be able to either store and later access their luggage quickly and easily while not slowing down passenger flow. A possible option of checked luggage for bigger items such as suitcases should be considered especially for smaller vehicles, where cabin space is further limited.

From a passenger perspective, it needs to be determined which kind of luggage handling is preferred by the passenger, being more independent and bringing all luggage into the cabin, or handing over larger items at the luggage check-in, potentially risking a waiting time during baggage claim, but having a more hassle-free experience during transit.

In terms of safety, all luggage that is brought into the cabin has to be stored safely in order to not cause a potential hazard in case of emergency braking. For safety reasons, passengers should not be encouraged to stand up accessing luggage during transit in order to limit risk of being unsecured in case of a sudden emergency braking event.

Three options have been considered and their pros and cons are presented in the table below.

Concept	Pros	Cons
All luggage allowed on board, similar to rail.	Passenger has luggage access at all times No extra service required	Increased vehicle size or reduced vehicle capacity Increased embarking/disembarking time and station footprint Uncomfortable to travel with large luggage
Personal bag and carryon are allowed on board. Large luggage is checked in. Similar to aviation.	Embarking/disembarking can be quick It is more comfortable to travel with small luggage.	A separate service is required to be implemented to handle luggage (Much like airports), resulting in increased costs Door to door trip time is increased, checking in luggage adds time. Increased station footprint
Personal bag and carryon are allowed on board. Large luggage is sent via an end to end, separate service	Embarking/disembarking can be quick It is more comfortable to travel with small luggage.	A separate service is required to be implemented to handle luggage resulting in increased costs Luggage and travel times can be different

It is difficult to come to a conclusion regarding luggage, it is recommended that a study is done to further elaborate on this aspect. As of now, the third option is recommended, it is perceived to provide the best travel experience and does not affect the station footprint.

5.3.4.2 SEATS

Seats are considered as one of the most important components of any interior as they have a significant impact on passenger experience and satisfaction. Seat design is also crucial for economy of space and weight, and also play a vital role in passenger safety.

Regarding seating, several factors should be considered:

Restraints	For safety reasons, regarding HARDT's projected emergency deceleration performance, seats must be equipped with seatbelts. Potentially, in case wearing a seatbelt is mandatory, the seat should be further equipped with a seat occupation and seatbelt buckle sensor, which could indicate whether or not a seatbelt is applied correctly.
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Width	Minimum of 55 cm. This is based on Hardt's previous work, and it is considered as the minimum comfortable with for a growing population. A more detailed study is recommended. See chapter 5.2.15.2.1 for reference sizes. The generous width is possible in hyperloop since it does not share the stringent weight requirements of aviation.
Pitch	Minimum of 105 cm. This is based on Hardt's previous work, an anthropometric study based on Dutch P99 male model. The generous pitch is possible in hyperloop since it does not share the stringent weight requirements of aviation.
Orientation	Seats can be arranged facing a single direction or facing each other in groups. A detailed study on passenger demand can help finding an appropriate relation of forward-facing rows and opposed group seating configurations. The option of swiveling seats, as used on e.g., Japanese Shinkansen HSR, has been considered as too complex and heavy, among other complications. A seat slightly rotated around its vertical axis can help creating more leg/(feet)room in more narrow interiors which feature a curved sidewall which tucks in towards the floor.
Anthropometrics	These measurements will be used as a basis for the overall interior packaging concept. Depending on the relevance, P99/P5 Male/Female measurements will be used to develop a concept that is suitable for the largest amount of people possible.
Features	May feature an adjustable component, such as backrest, headrest, armrests and/or footrest. Also, data and power outlets, coat hooks and folding tray tables with cup holders. Furthermore, seats may require to be marked and offer digital reservation signage. Regarding entertainment, a BYOD (Bring Your Own Device) policy may be implemented.

HARDT Hyperloop has secured support from RECARO, whose RECARO Aircraft Seating Division is a market leader in aviation seating and provides aircraft seats in all categories to renowned airlines all over the world.

5.3.4.3 ENVIRONMENTAL CONTROL UNITS

Usually located above the seat in the overhead bin, aircraft cabins feature PSUs (Personal Service Units). These typically include a reading light, an adjustable air vent nozzle, a flight attendant call button, a speaker, signal lights such as the outdated non-smoking sign and the seatbelt sign, as well as the hidden emergency oxygen mask, a feature which is also potentially required in the hyperloop application.

5.3.5 **CONCEPT TRADEOFF**

5.3.5.1 **MODAL COMBAT**

A tradeoff process was set up in which Hyperloop was compared to other modes of transport based on the travel characteristics. The reasoning behind this process can be found as part of chapter **Error! Reference source not found.**, in which the ideal market-fit for hyperloop is studied.

In this evaluation method, Hyperloop is directly compared against rail and aviation. A mark is placed on a scale from -3 to +3, favoring either one mode of transport or the other. A tie is also possible, resulting in 0 points for both contenders. In preparation for this method, the following criteria were bound together into service characteristics and safety is assumed as a given.

Characteristic	Criteria	Definition
Comfort	Vehicle environment	Climate control, noise, seat comfort, space, light, etc.
	Availability of seats	Balance between available seats, and ability to book
	Vehicle facilities	Sanitary, luggage and PRM facilities
Convenience	Frequency of service	How often vehicles depart/arrive per route
	Ease / Ease / number of transfers	How easy/how many transfers are expected in a trip
	Embark / disembark time	How quickly the vehicle is emptied and onboarded
Operations	Reliability / punctuality	How well does the system perform against expectations
	Transit time	How long should passengers remain in the vehicles
Costs	Ticket price	Cost of a single ticket
Sustainability	Energy usage	How much energy is required per trip
	CO2 footprint	How much CO2 is produced per trip
	Future prospect	How is the mode of transport expected to improve
Perceived safety	Perceived safety	How safe is a mode of transport as per public view
Door to door	Door to door trip time	How long is the entire trip, including first mile, embarking, transit time, disembarking, and last mile

Figure 31. Overview of service characteristics of modal combat

Hyperloop was divided in the three scenarios for this comparison, S, M and L. Two other points of comparison were added, the length of the trip. The overall comparison was then Rail, Hyperloop S, M, L and aviation during short and long trips. Furthermore, weights were assigned to the different service characteristics based on literature. (Gov.uk, 2021)

The tradeoff process was carried out by means of group discussions, in which the scores changed multiple times, and modifications to the model were made. Three entire versions of this Modal Combat model were made, until the work group was satisfied with the process and the results appear to be fair.

It is, however, important to note, that even though the comparison has been done as fair as possible, there is real data for aviation and rail in which the grades can be based. This is not the same for hyperloop, however, there is no option in this regard.

5.3.5.1.1 OUTCOME

The outcome of the modal combat was favorable to all hyperloop concepts. Scoring highly above the rest of the modes and very close to each other. The scores are presented in the table below. Notable is the high score of regional rails in short distance trips, this is mostly due to the good performance on short trips.

	Hyper S	Hyper M	Hyper L	Regional rail	Intercity rail	Aviation
Weighted total	124	126	123	110	98	90
Total	21	21	21	19	16	13

Figure 32. Modal combat scores overview

The full chart can be found in Appendix C, chapter 8.3

5.3.5.2 HYPER COMBAT

After the hyperloop was compared to other modes, the trade-off process continued towards the comparison between the different hyperloop concepts, including paradise loop, disregarding the other modes. The different scenarios were compared in the following categories.

Comfort	Sanitary systems Luggage handling Personal space Perceived personal space On board services / amenities Seating comfort Privacy
Flexibility	Booking / reserving Departure frequency Flexible scheduling
Trip times	(Dis)embarking time Luggage handling Trip time Ease of transfers Number of transfers
Accessibility	PRM friendliness
Reliability	Weather sensitivity Congestions Accidents Disruption prone Delays
Implementation	Space requirements Station size

	Ease of implementation NIMBY
Costs	On board services Safety systems implementation Ticket cost Infrastructure cost Cost per km travelled
Sustainability	Energy usage CO2 footprint Future prospect
Throughput	Throughput
Safety	Evacuation system complexity

Figure 33. Overview of characteristics of hyper-combat comparison

The work group carried out the process during several discussion sessions.

5.3.5.2.1 OUTCOME

In order to present a clearer idea of what the results mean, after the totals were achieved, the results were re-mapped to a scale of 0-100, in which the 100 represents the perfect hyperloop, or the Paradiseloop. The results are the following. The full chart can be found in Appendix A, chapter 8.1

	Hyper S	Hyper M	Hyper L	Paradiseloop
Total	93	113	111	130
Total adjusted to 100	72	87	86	100

Figure 34. Hyper combat score overview

5.3.5.3 PERSONAS

Working with different target groups, based on factors such as mindset, culture, income, age or nature of travel, different personas can be created that represent individual examples of those target groups. They differ in the weighting / multiplier of the above-mentioned variables: Whereas for one persona a short travel time is the most important factor, others may focus on convenience or low-ticket costs. The multipliers can be adjusted accordingly, altering the result for each mode and therefore eventually the individual ranking within this comparison.

The results are the following. The full chart can be found in Appendix A, chapter 8.1

Scenario log	Hyper S		Hyper M		Hyper L		Paradiseloop	
	Weighted by service ch.	Weighted by s.ch & criteria	Weighted by service ch.	Weighted by s.ch & criteria	Weighted by service ch.	Weighted by s.ch & criteria	Weighted by service ch.	Weighted by s.ch & criteria
Design team	70	68	86	87	86	87	100	100
Politician	71	72	86	87	83	83	100	100
Investor/bank	70	71	84	85	83	85	100	100
Passenger	72	72	87	88	86	86	100	100
HL operator	69	66	86	87	89	93	100	100

5.3.6 TRADE-OFF CONCLUSION

Hyperloop M and L score almost the same points. They offer similar levels of comfort, accessibility, reliability, etc. The main differences are in where the costs are distributed. Hyperloop L is more costly and difficult to implement, however it is expected to be cheaper in operation. Hyperloop M is cheaper and easy to implement but may be a bit more expensive to operate. They can both handle the required throughput and provide similar levels of experience and safety, which is where Hyperloop S fails.

Furthermore, when accounting for cargo integration, meaning that hyperloop needs to transport passengers and cargo, the jump between Hyperloop M and L is not very large and it is considered unnecessary when considering palletized cargo transport. However, Hyperloop L would be capable of transporting air cargo of the ULD L39 (Biggest available), while Hyperloop M would have to settle for ULD L29 or smaller (One size smaller than L39). For the full overview refer to chapter 5.3.3.5

It is then the recommendation to continue development with Hyperloop M as a baseline, on the basis of overall lower implementation costs and complexity, which is one of the big challenges of hyperloop implementation.

6 CONCEPT DESCRIPTION

This section compiles the learnings from the studies about passenger S, M, L into one recommended concept. It is based on the Hyperloop M, denominated M_EVO.

6.1 SERVICE CHARACTERISTICS

The market and safety requirements have been thoroughly discussed during the tradeoff process. They are presented in the following section and paired with the parameter that they define and/or affect. For more information refer to chapter 5.

Service characteristic	Requirement	Parameter
Throughput	20,000 passengers per hour per direction	Capacity of 42 passengers including PRMs, with possibility for platooning.
Convenience	Frequent departures/arrivals, easy/few transfers, short egress/access time	Vehicles feature two large doors at both sides for quick embarking/disembarking.
Reliability / Punctuality	More reliable than rail and aviation.	Immune to weather and track level crossings Fully automated system reduces human error
Low transit time	Access times, trip time, transfer time.	Low access and trip times as well as few transfers make it the best alternative for trips between 25-2500km
Vehicle facilities	Include PRM, passenger restrains and lavatory access	PRM can embark/disembark on their own during normal and emergency operation. Seatbelts and lavatories are present in every hyperloop vehicle.

Figure 35. Overview of Service characteristics

6.2 CONCEPT HYPERLOOP SERVICE

In this section, an overview of the hyperloop service is presented.

6.2.1 PASSENGER JOURNEY

An overview of the passenger journey is presented below.



Figure 36. Passenger journey

The following table presents how each of the touchpoints affects the overall service characteristics and their requirements. These tie directly to the goals of the document.

Touchpoint	Description
Flexible ticketing	Improves convenience. Hyperloop ticketing works as a mix between rail and aviation. Tickets can be reserved to ensure a place in a trip. However, they can also be sold using a subscription that allows on-the-fly boarding. Depending on availability.
Easy access to hub	Improves convenience and transit time. Hubs are to be multimodal and integrated as close as possible to the city centers. It is expected that some hubs will be outside the city centers, but not as far as airports. This improves the connectivity of the hubs.
Right of travel at walking pace	Improves convenience and transit time. The passenger identification, ticket/subscription check and security check should be as seamless as possible. The goal is to have these three processes occur concurrently at a walking pace. This means, the passenger should not be required to stop.
Short embarking time	Improves convenience and transit time. Just like in regional rail, the embarking should be quick and efficient. The vehicles are small and feature large openings to allow this to happen. This positively impacts the total duration of the journey and the departure/arrival frequency.
Short trip	N/A
Short disembarking time	Improves convenience and transit time.



	The disembarking should be quick and efficient. The vehicles are small and feature large openings to allow this to happen. This positively impacts the total duration of the journey and the departure/arrival frequency.
Connected exit / short transfer	Improves convenience. Hubs are to be multimodal and integrated as close as possible to the city centers. Transferring to other mode of transport for the continuation of the trip or last mile should be seamless.

Figure 37. Overview of passenger touchpoints

6.2.2 CONCEPT OF OPERATIONS DIAGRAM

The following diagram presents the hyperloop concept of operations, in which the passenger and cargo activities are described.

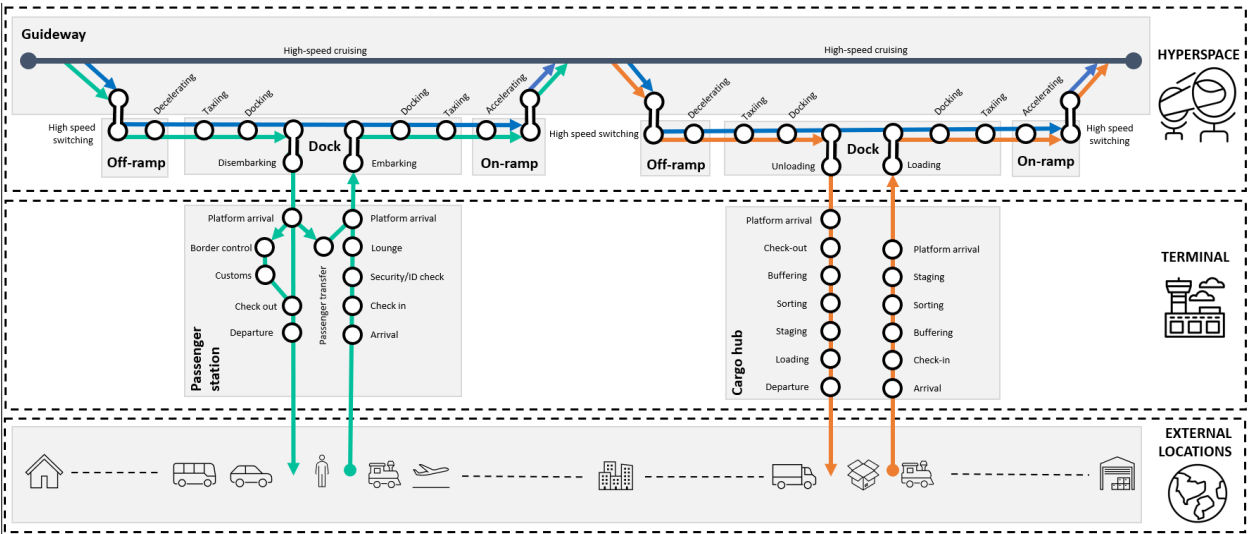


Figure 38. Concept of operations, courtesy of Hardt

Note that both cargo and passenger vehicles coexist in the same infrastructure. And the large luggage transport happens as a separate process. Furthermore, PRMs do not require additional, therefore, they don't show on the diagram.

6.3 CONCEPT HYPERLOOP PRODUCT

6.3.1 VEHICLE CONCEPT

The M_EVO is based on the M-sized variants, but with a slightly larger inner cabin width of 2.4 m and a resulting tube diameter of 3.5m.

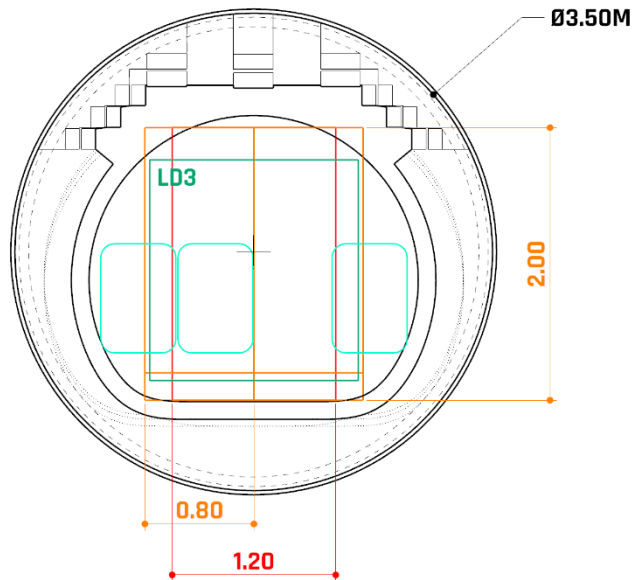


Figure 39. Hyperloop M_EVO cross section, indicating seats and cargo capacity

The length of the vehicle remains undefined, pending vehicle packaging in later stages, in particular the packaging of subsystems and magnets.

Like M, it uses a 2+1 seating arrangement for the standard cabin, as well as a spacious 1+1 arrangement for a business class section. Furthermore, it accommodates cargo, according to the set requirements.

The LOPA is mirrored across the length of the vehicle, which ensures a consistent travelling experience every time, as the vehicle travels in both directions. In its current form, at a cabin length of 25 m, the vehicle seats 42 passengers: 24 in the standard cabin and 12 in two business class

sections. From the remaining 6 seats, those in the front section in the direction of travel (3) can be reserved for passengers in wheelchairs and their companions, the remaining three at the end of the vehicle in direction of travel can be given to standard passengers. This ensures that a passenger in a wheelchair can access the front emergency exit easily and safely.



Figure 40. Hyperloop M EVO side view.



Figure 41. Hyperloop M EVO top view. Note the PRM accommodation indications.



The PRM access has also been revised in this update, with ingress / egress times in mind. After entering the vehicle through the door, the passenger stream is divided into two parts, on each side of the door. At front door in the direction of travel, Premium passengers and PRM / wheelchair users turn left, while regular passengers turn right into the cabin center section. This allows the PRM /wheelchair users more time to transfer to their seat and safely store their wheelchair.

Furthermore, emergency exits have been located at the nose and the tail of the vehicle, and PRM are able to exit with autonomy. The business section leaves a wider aisle through which the PRMs can exit on their own.

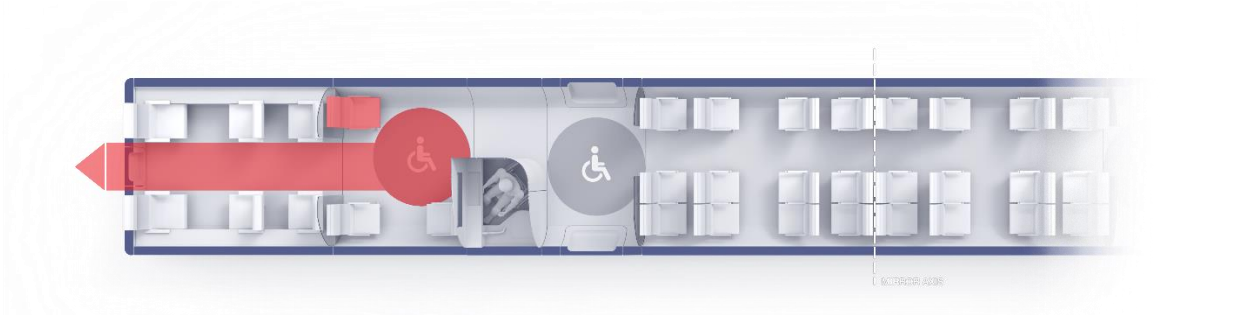


Figure 42. PRM emergency exit path

The lavatory has been enlarged in order to accommodate a passenger sitting in a wheelchair. The design of the lavatory is inspired by the LAV4ALL research groups’ work shown at <https://www.lav4all.facc.com>.

6.3.2 SAFETY CONCEPT

The safety concept is described in detail in chapter 5.2.4. It is comprised of the following aspects.

Aspect	Description
“Brick wall” scenario	The distance between vehicles traveling in each hyperloop line must be such that in case of an emergency, all other vehicles, with the exception of the vehicle involved in the emergency can safely come to a full stop and have access to emergency exits.
Pressure control	Vehicle and infrastructure pressure control is instrumental in the safety of the system. In case of emergency, the tube can be brought to ambient pressure in a matter of seconds, and emergency oxygen can be deployed in case of a pressure loss in the cabin. Any combination of these two capabilities is possible.
Emergency evacuation	Vehicles and infrastructure feature emergency exits that allow all passengers, including PRMs to autonomously and safely exit from the hyperloop infrastructure.
Passenger restrains	All passengers and their carryon luggage must be restrained during operation with the exception of lavatory visitation.

Figure 43. Overview of safety aspects

7 RECOMMENDATIONS

7.1 SCHWEIZER DESIGN

Go big!

Make the tube as large as possible.

The only limiting / negative / restrictive factor speaking against a large tube diameter is construction cost and possibly energy cost for evacuation. All other factors clearly favor a large tube diameter: Capacity. Passenger Comfort. PRM accommodation. Flexibility & Versatility. Growth opportunities. Turning Radius & Track Layout. Differentiation from Aviation. Competitiveness with HSR.

Cost per passenger drops with size even if initial investment / acquisition costs are higher. Less doors / emergency exits / lavatories / staff / ... per passenger.

We think that the advantages of Hyperloop allow for ticket prices to be marginally higher than for competitive modes of transport and it is still acceptable. Same must be valid from the investor / operator point of view. If the benefit is right, it's ok to be more expensive.

Those who believe in Hyperloop want it to be as big as possible. If you believe in its success, you believe in its growth. Since the system is not up scalable, you have to start with the biggest version possible.

As an example: The Stuttgart S21 train station. A critical point of criticism has been and still is, that it will be too small for the existing demand the day it opens, requiring costly band-aids which are already being developed now, before completion of the main project.

Aviation has an advantage here; it is highly flexible and up scalable. Higher demand? Fly more often. Even more demand? Buy (or lease) bigger aircraft. Even more demand? Extend terminal / number of runways. The sky's literally the limit, but Hyperloops bottleneck is not the station, it is the fixed track and its limited capacity.

For us, it doesn't matter how big the tube is, as long as it is the biggest one you were able to get. We'll take it from there and turn it into the best result within those limitations.

Downscaling is always possible. A shorter, more exclusive vehicle, 'the private jet', is possible, and a long high-capacity mass transport, too. Depending on future developments (decades ahead) it can adjust and is future proof.

The biggest tube is the most reasonable.

7.2 HARDT

It is very difficult to balance all of the different ingredients of this recipe. With the mindset of thinking that this mode of transportation could last half a century or more before another breakthrough takes its place.

We've built this based on a lot of assumptions, but ultimately these assumptions reflect what we believe the market needs, not only today but for years to come. To know this, with certainty, is not possible. For me it boils down to building a system that will move the highest amount of people, therefore produce the most benefit, for as many years as possible. In the end, the limiting factor will be the tube diameter.

Simply put, we should aim for the largest possible system. The two main opposing forces would be costs and public acceptance. However, we should not build anything smaller than Hyperloop M, because it makes no sense to have a modality like this.



8 APPENDIX

8.1 APPENDIX A. HYPER COMBAT

Hyper combat												
To be scored from 1 to 5												
Service characteristic	Weight	Criteria	Hyper S		Hyper M		Hyper L		Paradiseloop			
Persona	Passenger		Weighted by service ch.	Weighted by s.ch & criteria	Weighted by service ch.	Weighted by s.ch & criteria	Weighted by service ch.	Weighted by s.ch & criteria	Weighted by service ch.	Weighted by s.ch & criteria		
	30											
Comfort	4,0	5 Sanitary systems	4	1,0	4,0	4,0	16,0	5,0	20,0	5,0	20,0	
		3 Luggage handling	2,4	1,0	2,4	3,0	7,2	5,0	12,0	5,0	12,0	
		2 Personal space	1,6	3,0	4,8	4,0	6,4	5,0	8,0	5,0	8,0	
		5 Percieved personal space	4	1,0	4,0	3,0	12,0	5,0	20,0	5,0	20,0	
		2 On board services / ammenities	1,6	4,0	6,4	5,0	8,0	5,0	8,0	5,0	8,0	
		2 Seating comfort	1,6	4,0	6,4	5,0	8,0	5,0	8,0	5,0	8,0	
		2 Privacy	1,6	5,0	8,0	4,0	6,4	3,0	4,8	5,0	8,0	
		4 Weighted average		10,9	5,1	16,0	9,1	18,9	11,5	20,0	12,0	
Flexibility	4,0	3 Booking / reserving	2,4	3,0	7,2	5,0	12,0	5,0	12,0	5,0	12,0	
		5 Departure frequency	4	5,0	20,0	4,0	16,0	3,0	12,0	5,0	20,0	
		3 Flexible scheduling	2,4	5,0	12,0	4,0	9,6	3,0	7,2	5,0	12,0	
		4 Weighted average		17,3	13,1	17,3	12,5	14,7	10,4	20,0	14,7	
Trip times	5	3 (Dis)embarking time	3	3,0	9,0	5,0	15,0	5,0	15	5,0	15	
		3 Luggage handling	3	3,0	9,0	5,0	15,0	5,0	15	5,0	15	
		5 Trip time	5	5,0	25,0	4,0	20,0	3,0	15	5,0	25	
		5 Ease of transfers	5	3,0	15,0	5,0	25,0	5,0	25	5,0	25	
		5 Amount of transfers	5	5,0	25,0	3,0	15,0	1,0	5	5,0	25	
		5 Weighted average		19,0	16,6	22,0	18,0	19,0	15,0	25,0	21,0	
Accessibility	3	5 PRM friendliness	3	1,0	3,0	5,0	15,0	5,0	15	5,0	15	
		3 Weighted average		3,0	3,0	15,0	15,0	15,0	15,0	15,0	15,0	
Reliability	5	3 Weather sensitivity	3	5,0	15,0	5,0	15,0	5,0	15,0	5,0	15,0	
		5 Congestions	5	3,0	15,0	4,0	20,0	5,0	25,0	5,0	25,0	
		5 Accidents	5	5,0	25,0	5,0	25,0	5,0	25,0	5,0	25,0	
		5 Disruption prone	5	5,0	25,0	5,0	25,0	5,0	25,0	5,0	25,0	
		5 Delays	5	3,0	15,0	4,0	20,0	5,0	25,0	5,0	25,0	
		5 Weighted average		21,0	19,0	23,0	21,0	25,0	23,0	25,0	23,0	
Implementation	1	1 Space requirements	0,2	5,0	1,0	4,0	0,8	3,0	0,6	5,0	1	
		2 Station size	0,4	5,0	2,0	5,0	2,0	5,0	2	5,0	2	
		1 Ease of implementation	0,2	5,0	1,0	3,0	0,6	1,0	0,2	5,0	1	
		4 NIMBY	0,8	3,0	2,4	2,0	1,6	1,0	0,8	5,0	4	
		1 Weighted average		4,5	1,6	3,5	1,3	2,5	0,9	5,0	2,0	
Costs	4,0	1 On board services	0,8	3,0	2,4	3,0	2,4	3,0	2,4	5,0	4,0	
		1 Safety systems implementation	0,8	1,0	0,8	4,0	3,2	5,0	4,0	5,0	4,0	
		5 Ticket cost	4	3,0	12,0	4,0	16,0	5,0	20,0	5,0	20,0	
		4 Infrastructure cost	3,2	5,0	16,0	3,0	9,6	1,0	3,2	5,0	16,0	
		1 Cost per km travelled	0,8	1,0	0,8	3,0	2,4	5,0	4,0	5,0	4,0	
		4 Weighted average		10,4	6,4	13,6	6,7	15,2	6,7	20,0	9,6	
Sustainability	3,0	3 Energy usage	1,8	5,0	9,0	5,0	9,0	5,0	9,0	5,0	9,0	
		5 CO2 footprint	3	5,0	15,0	5,0	15,0	5,0	15,0	5,0	15,0	
		1 Future prospect	0,6	5,0	3,0	5,0	3,0	5,0	3,0	5,0	3,0	
		3 Weighted average		15,0	9,0	15,0	9,0	15,0	9,0	15,0	9,0	
Throughput	1	5 Throughput	1	3,0	3,0	4,0	4,0	5,0	5,0	5,0	5,0	
		1 Weighted average		3,0	3,0	4,0	4,0	5,0	5,0	5,0	5,0	
Safety	1	Evacuation system complexity		1		5		5		5		
		Weighted average		1		5		5		5		
			Hyper S		Hyper M		Hyper L		Paradiseloop			
Grand total				93	72		113	88		111	85	
Grand total (Adjusted to 100)				72	71		87	87		86	85	
										100	100	

8.2 APPENDIX B. SHIPMENT UNIT DISTRIBUTION

The results of the analysis of physical dimensions of shipment units suggest that shipment unit length (SU length) ranges from 0.6m to 12m. About 78% of the units were less than 1.2m long, and 87% of the market uses shipment units less than 1.5m long (Figure 44Error! Reference source not found.). The mean value of SU length was 1.7 m with a mode and median at 1.2m.

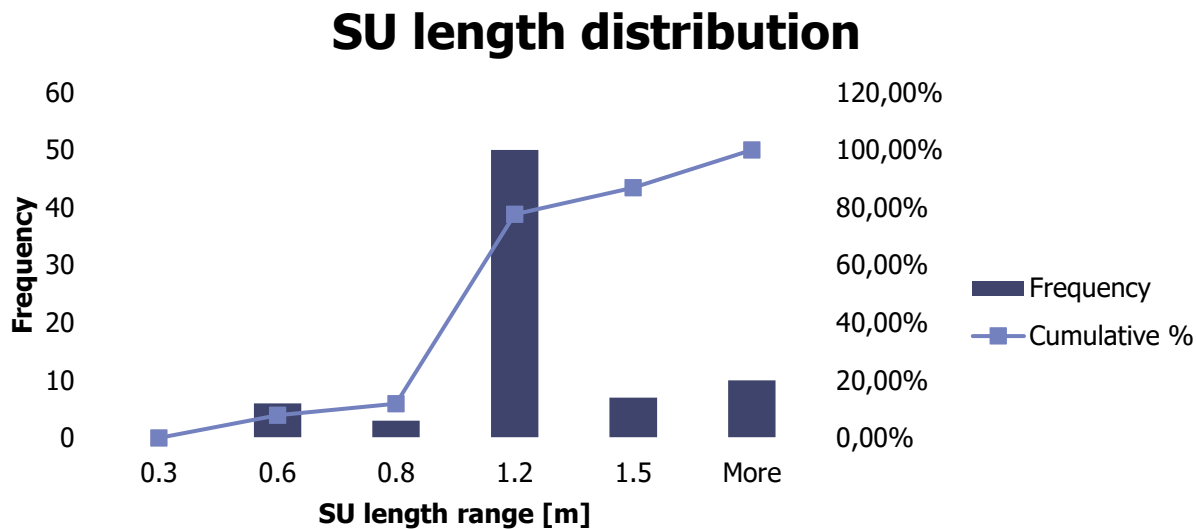


Figure 44: Shipment unit length distribution, sample = 76

Shipment unit width (SU width) ranges from 0.6m to 2.7m, with about 50% of the units being of less than 0.8m wide, and 90% being less than 1.2m wide (**Error! Reference source not found.****Error! Reference source not found.**). The mean value of SU width was 1m with a mode and median at 0.8m.

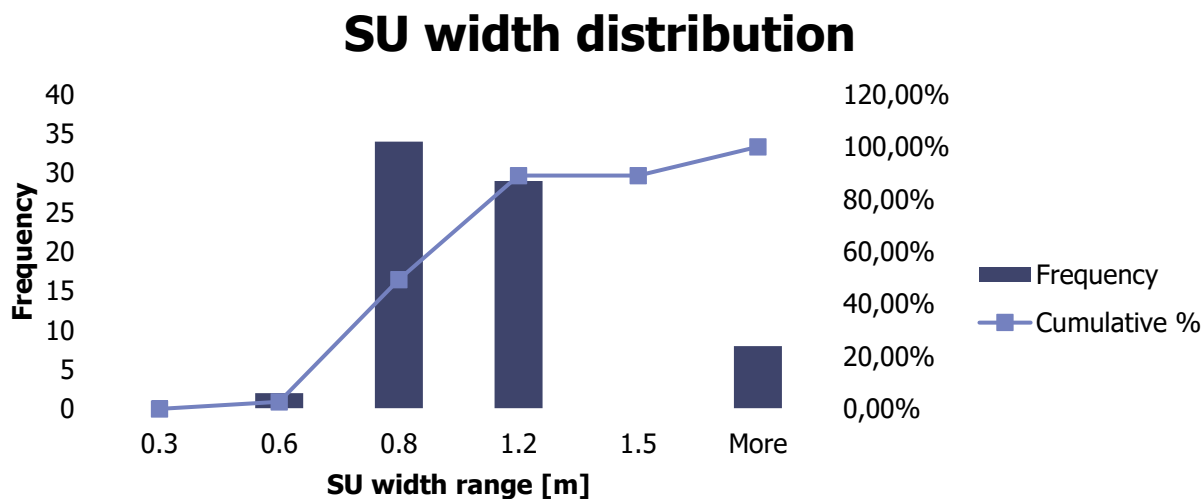


Figure 45: Shipment unit width distribution, sample = 73

In practical terms these results suggest that:

- about 36% of cargo is transported on euro pallets (1.2 x 0.8 m)
- about 31% of cargo is transported on block pallets (1.2 x 1.2 m)



In total, 67% of cargo is transported on shipments units of dimension below 1.2 x 1.2 m. This is the minimum the Cargo hyperloop needs to achieve. Expanding the possible length of shipment units to 1.5 m would allow for capturing additional 10% of the market up to 77%.

Shipment unit height (SU height) ranges from 0.7m to 2.7m, with about 49% of the units being of less than 1.2m high, and 92% being less than 2m height (Figure 46Error! Reference source not found.). The mean value of SU height was 1.35m with a mode at 1m and median at 1.25m.

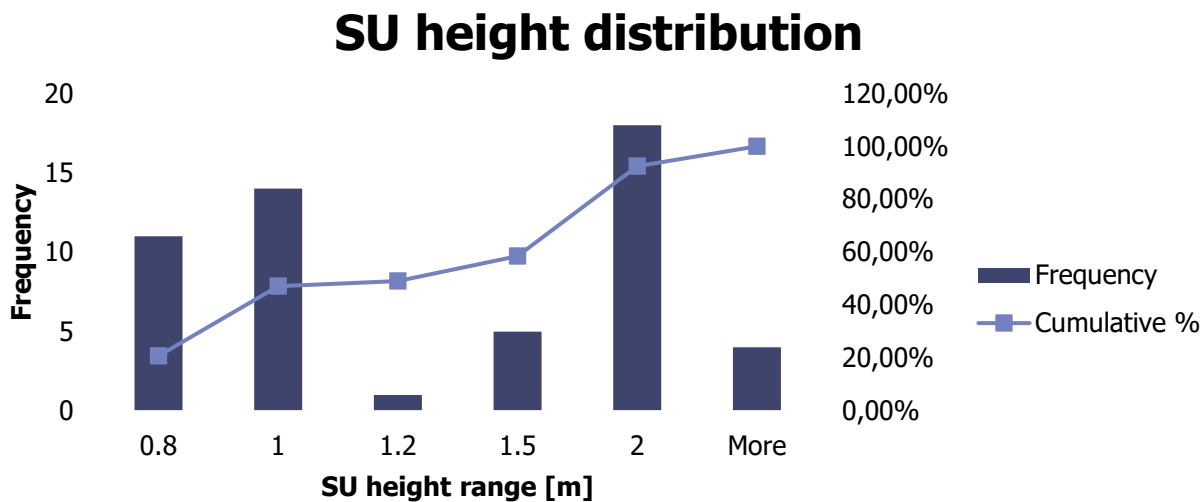


Figure 46: Shipment unit height distribution, sample = 53

8.3 APPENDIX C. MODAL COMBAT

Service characteristic		Criteria	Weight	Score	Weighted score	Score	Weighted score	Score	Weighted score	Score	Weighted score	Score	Weighted score	Score	Weighted score	Score	Weighted score	Benchmark (5 points)
					Hyper S		Hyper M		Hyper L		Regional distance		Rail		Rail		Aviation (Lufthansa economy)	
Comfort	1	Vehicle environment		1	3	3	10	3	2	2	4	4	2	2	2	2	10	First class HSR
	2	Availability of seats	✓	5	5	5	10	5	3	6	3	6	5	3	6	5	10	Plane / First class HSR
	1	Vehicle facilities	✓	3	5	5	5	5	3	3	4	4	4	5	5	5	5	Plane
Convenience	3	Frequency of service		5	5	5	15	5	4	9	3	9	9	3	9	3	3	Hyperloop
	2	Ease / amount of transfers	✓	10	8	4	10	8	2	4	2	4	4	3	4	3	6	Hyperloop 5
	3	Embark / disembark time	✓	4	12	4	12	4	4	12	5	15	12	5	15	1	1	Regional rail
Operations	4	Reliability / punctuality	✓	4	16	4	16	4	16	4	16	16	4	16	4	4	4	16
	4	Transit time	✓	4	5	5	20	5	4	4	1	4	4	3	4	3	12	
	4	Ticket price		4	16	4	16	4	4	5	5	20	20	5	20	5	20	
Sustainability	1	Energy usage		5	5	5	5	5	5	4	4	4	4	4	4	1	1	Refer to energy chart
	1	CO2 footprint		5	5	5	5	5	5	4	4	4	4	4	4	1	1	Refer to energy chart
	1	Future prospect		5	5	5	5	5	5	2	2	2	2	2	2	3	3	
Perceived safety	1	Perceived safety		1	1	1	1	1	1	5	5	5	5	5	5	5	5	
Door to door	1	Door to door trip time	✓	5	5	5	5	5	5	4	1	1	3	3	3	3	3	Refer to Trip table
Total - Weighted				103		126		123		110		98		90		13		
Total - Not weighted				21		21		21		19		16		13				

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